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Measuring the Unmeasured: A Process-Level Investigation of Scrap Generation in MTO Sheet Metal Manufacturing

PAPER WITHIN: Industrial Product Realisation

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This final project work has been carried out at the School of Engineering at Jönköping University in the subject area Industrial Product Realization. The work is carried out within the programme Master of Science in Engineering Industrial Product Realisation. The authors take full responsibility for opinions, conclusions and findings presented.

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Abstract

Purpose – The purpose of this study was to apply the Define, Measure, Analyse, Improve, Control (DMAIC) framework as a structured approach for identifying and analysing scrap generation at the process-level in make-to-order (MTO) sheet metal production, and to evaluate how process-level measurement can serve as a foundation for structured scrap reduction.

Methodology – A single-site case study conducted at an air handling unit (AHU) manufacturer in southern Sweden, using a concurrent mixed-method approach, combining structured observations, semi-structured interviews, enterprise resource planning (ERP) extraction and manual scrap bin weighing across the DMAIC phases, with the Control phase excluded due to the time constraints.

Findings – A combined ERP extraction and manual weighing approach successfully established a process-level scrap baseline where none previously existed. The punching stage was the dominant scrap contributor, with coil width mismatch as the primary root cause. A significant measurement gap was identified at the blank removal station, which contributed to 19% of total weighed scrap despite not being visible in the ERP data. Four improvement proposals were developed spanning kaizen- and kaikaku-type interventions.

Implications – The study demonstrates that process-level scrap measurement is achievable low-automation MTO environments when the measurement approach is designed around actual data availability. It contributes to the applied DMAIC literature by showing that the Measure phase must treat measurement system design as an active output rather than a precondition.

Limitations – Single case study conducted over one academic semester. Manual weighing carried human-dependent variability, and proposed improvements were not implemented or validated.

Keywords: *DMAIC, MTO, material scrap, process-level measurement, scrap baseline, sheet metal manufacturing*

List of Abbreviations

AHU	Air Handling Unit
ATO	Assemble-to-Order
CoQ	Cost of Quality
DMAIC	Define, Measure, Analyse, Improve and Control
ERP	Enterprise Resource Planning
ETO	Engineer-to-Order
KPI	Key Performance Indicator
LoA	Level of Automation
MSA	Measurement System Analysis
MTO	Make-to-Order
MTS	Make-to-Stock
NC	Numerically Controlled
OEE	Overall Equipment Effectiveness
OPP	Order Penetration Point
PAF	Prevention-Appraisal-Failure
PDCA	Plan-Do-Check-Act
RQ	Research Question
SIPOC	Suppliers, Inputs, Process, Outputs, Customers
UTL	Material Utilisation
VSM	Value Stream Mapping
WIP	Work-in-Progress

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1 Introduction

This chapter introduces the study by establishing the industrial and production context for material scrap in sheet metal manufacturing. It presents the research gap and problem description that motivated the study, followed by the purpose and research questions that guided the study. The chapter concludes with the delimitations and an outline of the thesis.

1.1 Background

The European Union has committed to reducing net greenhouse gas emissions by at least 55% by 2030 compared with 1990 levels, placing increasing pressure on industry to improve both energy and material efficiency (European Commission, 2020). Manufacturing plays a central role in the European economy, but it also contributes substantially to total emissions, resource use, and the generation of material scrap (European Commission, n.d). In this context, improving the use of material in manufacturing is not only an economic concern but also part of broader environmental and climate objectives (Shahbazi et al., 2016).

This issue is particularly important in metal-intensive production, where a significant share of input material never becomes part of the final product. In the steel industry, at least 26 % of liquid steel never makes it into finished products, instead being lost as material scrap (Milford et al., 2011). Reducing it could cut CO_2 emissions by approximately 16 % in the steel industry. While these findings are presented at the level of the steel supply chain, they point to a broader issue that is also relevant for sheet metal manufacturing. In sheet metal manufacturing, material scrap can arise at several points across a multi-step production system, accumulating through sequential process steps such as coil cutting, punching, and forming, making scrap reduction important from both sustainability and productivity perspectives (Ingarao, 2017; Smirnov et al., 2019).

This is closely related to the concept of material efficiency, which refers to how much of the raw material input is converted into useful product output (Shahbazi et al., 2016). In sheet metal manufacturing, it is strongly affected by how material is used in operations such as coil cutting, punching, and forming, where material scrap may arise from trimming, nesting decisions, part geometry, process settings, and quality deviations (Ingarao, 2017; Smirnov et al., 2019). Improving material efficiency therefore requires understanding not only how much scrap is generated in total, but which operations and conditions generate it.

Despite its importance, material scrap is often monitored in ways that provide limited insight into where losses actually occur. In many manufacturing settings, scrap is reported in aggregate, for example as a plant-level or line-level figure, which shows the overall scale of material scrap but says little about which operations generate it (Kang et al., 2016). Similarly, commonly used performance measures such as Overall Equipment Effectiveness are mainly designed to capture availability, performance, and quality losses at equipment or line-level rather than process-specific material utilisation (UTL) (Domingo & Aguado, 2015). When scrap is only visible in aggregated form, the production system becomes difficult to interpret, since the total amount of scrap may be known while the operations, product variants, or production conditions behind it remain unclear.

Production systems vary in terms of volume, variety, and flow configuration. There is a range of different flow configurations from project flow to continuous processing. Hybrid systems that combine batch-oriented and line-based stages are common in practice (Bellgran & Säfsten, 2010). In such

systems, scrap volumes may be influenced not only by the performance of individual operations, but also by earlier decisions related to how products are grouped, sequenced, or prepared before entering later stages.

Production strategy adds a further layer of complexity. In make-to-stock environments, production is typically planned against forecasted demand under relatively stable conditions, whereas make-to-order (MTO) environments are characterised by variation in product mix, order size, routing, and processing requirements between orders (Olhager, 2003). In sheet metal manufacturing, this means that cutting layouts, material quantities, and process conditions may change from one job to another, so differences in scrap levels do not necessarily reflect process problems alone but may also result from inherent variation between orders and product requirements (Mathew & Johansson, 2023; Soman et al., 2004).

Structured improvement methodologies have been developed to address exactly this type of problem. The Define, Measure, Analyse, Improve, Control (DMAIC) framework provides a data-driven structure for reducing process variation and waste by first establishing a clear measurement baseline before attempting to identify causes and implement improvements (Sörqvist & Höglund, 2017). This makes it particularly well suited to manufacturing contexts where the scale and sources of scrap generation are not yet fully understood, and where measurement must precede improvement. Applied to scrap reduction, DMAIC has demonstrated significant results in practice, with case studies showing waste reductions of up to 50% in manufacturing environments (Muhammad Hamad et al., 2021).

These challenges do not exist independently. In MTO production environments, the order penetration point (OPP) means that material specifications, batch sizes, and processing sequencing can vary from one order to the next (Olhager, 2003). As a result, scrap levels fluctuate between jobs for reasons completely unrelated to process performance. In hybrid production systems, where batch-based upstream stages feed into line-based downstream stages, configuration differences between segments mean that automation levels can vary across the production flow (Holweg et al., 2018). Where automation is low, the availability of digital production data is also low (Frohm et al., 2008). Where data is available, it is often reported at the plant or line level rather than at the individual process step (Kang et al., 2016). Together, these conditions make it difficult not only to measure scrap at the process-level, but also to distinguish between scrap that is systematically built into the process design, scrap that varies with order characteristics, and scrap arising from abnormal events such as machine failures or handling errors (Datar & Rajan, 2017; Juran, 1986). Without this distinction, improvement efforts lack an empirical foundation.

1.2 Research Gap

Structured improvement frameworks such as DMAIC provide documented approaches for identifying and reducing process variation, and their application to scrap reduction in manufacturing has been demonstrated across several industrial contexts (Enache et al., 2023; Muhammad Hamad et al., 2021). However, when more closely examining this existing literature, it reveals a consistent pattern: existing studies have mostly been conducted in environments where process-level measurement data was already available prior to the improvement effort and where production conditions were sufficiently stable to allow direct application of the framework. In these cases, the Measure phase of DMAIC served to quantify a known problem rather than to establish a measurement capability that did not exist prior to the study.

This precondition is frequently absent in practice. Shahbazi et al. (2016) identify the inability to link material scrap to specific operations as one of the most influential barriers to material efficiency improvement in manufacturing, noting that companies may be aware of total scrap volumes but still lack the process-level data needed to act on them. Santos et al. (2025) describe in a similar manner how the absence of structured scrap control at the process-level undermines improvement efforts in metal-mechanical production. Together, these findings suggest that many manufacturers lack the prerequisite for applying DMAIC and must establish a reliable measurement baseline as a part of the improvement effort.

This challenge is amplified in complex MTO production environments with hybrid process structures, where variation in order specifications and batch sizes means that scrap levels fluctuate between jobs for reasons unrelated to the performance of the process (Olhager, 2003; Soman et al., 2004). In such contexts, establishing a baseline measurement and measuring normal process behaviour is non-trivial, and to be able to distinguish between process-driven scrap and order-driven variation requires a measurement approach designed for that purpose. Existing DMAIC case studies do not address this combination of challenges: the simultaneous need to design a process-level measurement approach, account for MTO-driven variation and apply DMAIC across a multistage hybrid production system.

This thesis seeks to address that gap by applying DMAIC in an MTO sheet metal production context where process-level scrap measurement did not previously exist, treating the establishment of measurement capability as an integral part of the improvement effort rather than a given starting condition.

1.3 Problem Description

Scrap generation in manufacturing is commonly monitored at the plant or production line level, where aggregate figures form the basis of improvement initiatives. While this provides a broad picture of total scrap volumes that is comparable between production processes, it does not reveal where in the production process scrap is being generated, in what volumes, or for what reasons. As a result, improvement efforts are directed at the system rather than at the specific process steps that are responsible for the scrap generation. Without knowing the distribution of scrap generation within a production process, it is impossible to prioritise where to intervene or evaluate whether implemented changes have had any effect on scrap generation.

Gargeya (2005) notes that while managers rely on a broad range of performance measures, only a limited number connect directly to competitive priorities at the process-level. When scrap is reported only as a plant-wide aggregate, it can obscure how waste is distributed across the different stages and leaves improvement efforts without empirical foundation (Neely et al., 1995).

This challenge is especially pronounced in MTO environments, where order specific material specifications and batch sizes can substantially vary the amount of scrap being produced between jobs (Soman et al., 2004). The type of production system adds further complexity. In job shop environments, products follow variable routes through general-purpose equipment, making it difficult to consistently attribute scrap generation to specific process steps (Holweg et al., 2018). Even in line-based systems, the level of automation across process steps affects the availability of production data, as manual or semi-automated operations rarely generate the digital records needed to support a comprehensive process-level analysis (Frohm et al., 2008). Finally, material scrap itself is not a uniform phenomenon, normal spoilage, defect-driven scrap, and abnormal spoilage are fundamentally different in nature, yet

are often aggregated under a single figure, obscuring their contributions and generation (Datar & Rajan, 2017; Juran, 1986). These differences do not necessarily reflect process failures but can arise as a natural consequence of the MTO production strategy (Soman et al., 2004). Together, these conditions make it difficult to determine which operations generate most of the scrap, in what volumes, and for what reasons, leaving improvement efforts without an empirical foundation.

1.4 Purpose and Research Questions

Whilst frameworks exist for structured process improvement, approaches to measuring and categorising material scrap at the process-level remain inconsistent across industries, with no single standard methodology in widespread use (Schiffauerova & Thomson, 2006). This thesis seeks to address the gaps described in sections 1.2 and 1.3 by establishing a process-level measurement approach and applying DMAIC to identify and analyse scrap generation in an MTO sheet metal production context. The purpose of this study is to:

Apply DMAIC as a structured approach for identifying and analysing scrap generation at the process-level in an MTO sheet metal production environment, and to evaluate how process-level measurement can serve as a foundation for structured scrap reduction.

To address the purpose of this study, three research questions (RQs) have been formulated. Understanding how scrap can be identified and measured at the process-level is a necessary knowledge base for any structured improvement effort. Therefore, the first research question is:

RQ1: How can material scrap and UTL be defined and measured at process-level within an MTO sheet metal production system?

Establishing a measurement approach alone is insufficient if the factors that drive variation in scrap generation are not understood. In an MTO environment, the scrap levels are influenced by a range of factors related to order characteristics, production system properties and process behaviour. Identifying these sources of variation is essential for interpreting the collected data and determining where improvement efforts should be directed. Therefore, the second research question is:

RQ2: What challenges and sources of variation affect the measurement and control of material scrap in the scrap generating stages of an MTO sheet metal production system?

With measurement baseline established and sources of variation identified, the third research question addresses how this process-level data can be used to support structured improvement work. Since the Control phase falls outside the scope of this study, the focus lies in how measurement and analysis can inform the identification and proposal of targeted scrap reduction measures rather than their full implementation. Therefore, the third research question is:

RQ3: How can process-level measurement and analysis support the identification and proposal of scrap reduction measures?

1.5 Delimitations

This study is delimited to the production stages within a single manufacturing facility's insulated panel production line that generate scrap in the form of metal offcuts and sheet metal losses. Total material scrap was measured per production stage, however, without distinction between systematic normal

spoilage losses and abnormal spoilage within each stage, as physical separation at that level was not feasible. The analytical focus lies on systematic, material scrap arising as normal spoilage from coil cutting and punching operations. Abnormal spoilage arising from machine failures, material handling incidents, or operator errors is acknowledged as a contributor to total scrap volumes and discussed as a source of variation but is not the primary object of analysis. Production defects in the form of panels removed from the line due to quality failures were additionally identified as a potential scrap source during the Define phase. Unlike the abnormal spoilage categories above, which are excluded on analytical grounds, defect-driven scrap is excluded specifically because defect volumes could not be traced to individual process stages or linked to specific scrap quantities within the available data and is therefore acknowledged qualitatively but not quantified.

Other forms of waste produced by the production system such as rockwool insulation offcuts, plant energy consumption or time-based losses fall outside the scope of this study. Which specific production stages fall within this delimitation is determined through the data collection conducted in the Measure phase, the results of which are presented in section 4.

Regarding the DMAIC framework, the Control phase has been excluded from the scope of the study. Full implementation and long-term monitoring of the improvement measures suggested by the study requires a timeframe that extends beyond the duration of this thesis project. Recommendations relating to control and sustainability of the improvements are however addressed in the Future Research section.

1.6 Outline of the Thesis

Chapter 1 *Introduction*	Introduces the study by establishing the industrial and environmental context for material scrap in sheet metal manufacturing. It presents the research gap, problem description, purpose and research questions, and defines the scope and delimitations of the study.
Chapter 2 *Theoretical Framework*	Provides the theoretical framework for the study, which defines key concepts and provides the foundation for analysis and discussion chapters.
Chapter 3 *Methodology*	Provides the study's methodology, including the study design, research approach and data collection strategy. This chapter also addresses reliability and validity, ethical considerations and the use of artificial intelligence.
Chapter 4 *Results*	Presents the empirical findings of the study, structured according to the DMAIC framework.
Chapter 5 *Analysis*	Analyses the empirical findings in relation to the theoretical framework, structured around the DMAIC framework's Analyse and Improve phases.
Chapter 6 *Discussion*	Discusses the implications of the findings for theory and practice, reflects on the methodology and its limitations.
Chapter 7 *Conclusion*	Summarises the study's key findings by directly answering the RQ, assesses the contributions of the study and suggests future research.

2 Theoretical Framework

To clarify the structure of the theoretical framework, Figure 2.1 illustrates how the research questions are connected to the theoretical areas included in this chapter. The figure shows how the selected theories together define the study context and provide the basis for measuring material scrap, analysing variation, and identifying improvement opportunities in an MTO production system.

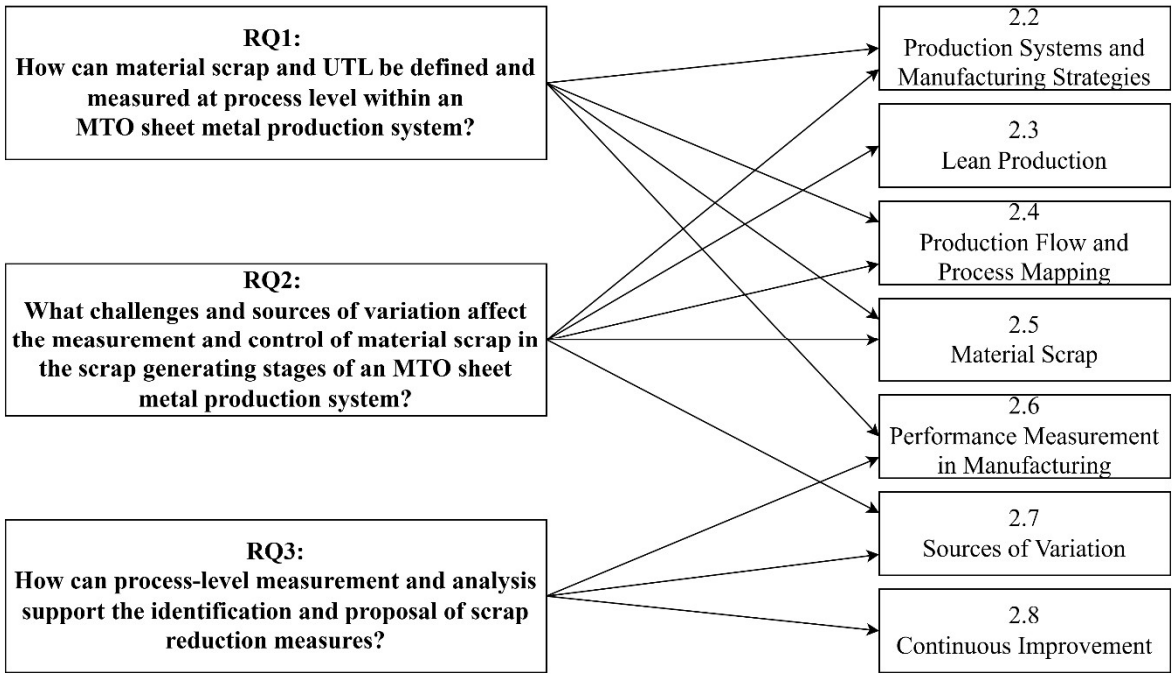


Figure 2.1
Connection between research questions and theoretical framework.

2.1 Production Systems and Manufacturing Strategy

A production system can be understood as a set of resources that transforms different inputs into finished outputs through a structured set of activities (Bellgran & Säfsten, 2010). In this context, materials, components, labour, information, and capital enter the system, move through a sequence of operations, and leave as products delivered to customers. The internal functioning of this input-output relationship is often illustrated using a simple “black-box” process model. In such a model, the detailed transformation steps are intentionally simplified in order to focus on what enters the system and what leaves it (Bellgran & Säfsten, 2010). From a more technical perspective, a production system can also be described using the concepts of process, operand, and operators. The system boundaries may be drawn at different levels, and everything outside these boundaries can be regarded as the external environment (Bellgran & Säfsten, 2010). This boundary perspective is essential when analysing a specific segment of a factory, as it provides theoretical justification for examining a bounded production process as a legitimate unit of study within a larger system.

2.1.1 Production Process

Production systems are typically categorised by the dimensions of volume and product variety, a relationship represented by the product-process matrix (Bellgran & Säfsten, 2010), see Figure 2.2. These two variables divide the production flow into five different flows which range from flexible, low-volume setups to rigid, high-volume continuous processes.

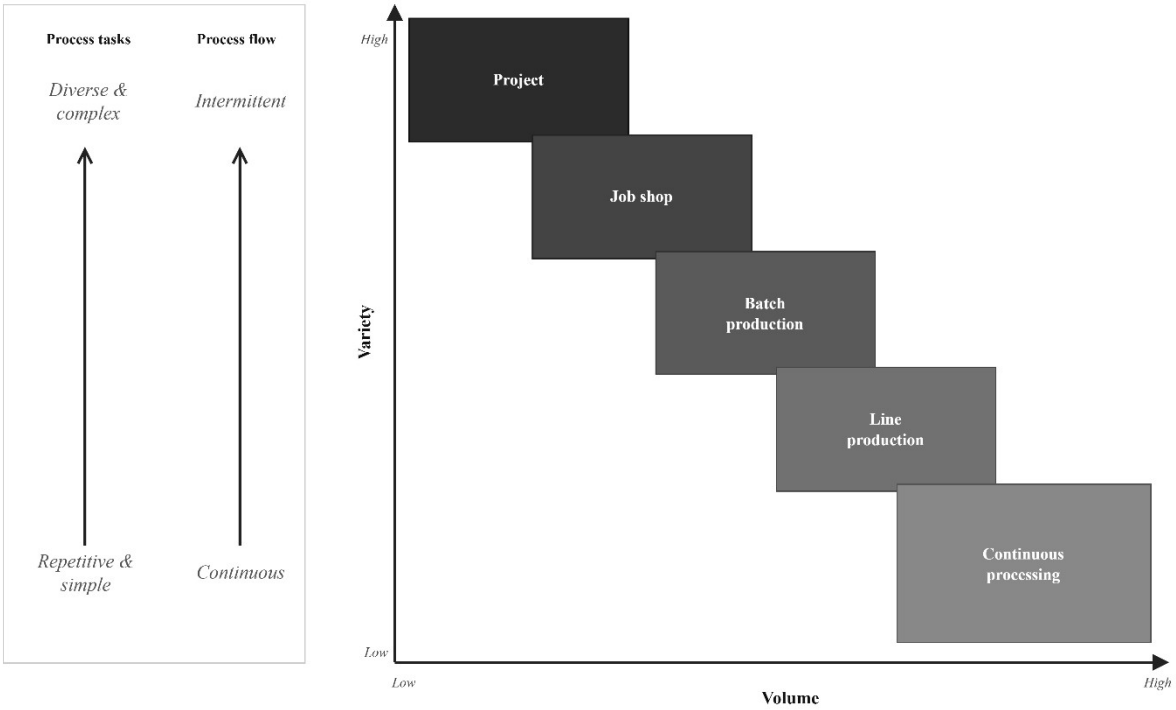


Figure 2.2
The spectrum of manufacturing processes: the product-process matrix.

Comment. Adapted from (Holweg et al., 2018)

Fixed position project flow is the type of flow whose main focus is customization (Holweg et al., 2018). Production of very large products, such as special products in small quantities, needs a production arrangement that instead of moving between different stations, material and personnel are being transported to the product at a certain location (Bellgran & Säfsten, 2010). The resources are often very generic in nature, which requires highly specialized workers, who need to execute a wide range of diverse tasks. The high customization level in this type of flow brings some operational challenges, which drive up the costs due to lower level of repetition and longer learning processes for workers.

Job-shop or functional flow is the most flexible process for creating a wide variety of products in bigger quantities (Holweg et al., 2018). This layout groups similar equipment together, where some of that equipment will be used heavily and some may not be used much at all. The material flow in this type of layout can be complex, which makes the routing of flow information essential in order to get the final product according to the product specifications.

Batch production is one more step towards continuous processing, with machines located in the direction of the flow instead of after functional similarity (Holweg et al., 2018). The similarities between job shop and batch production are many, with a few distinguishable differences which place this layout as a more standardized way for making products in large volumes and in many variants. To set the equipment up

in this flow, the target is to create a sequential flow for as many parts as possible. The main challenge in a batch production system is the grouping of the products, which can be done through comprehensive production flow analysis.

A line-based production system, frequently referred to as an assembly line or line flow layout, is a product-oriented configuration where manufacturing equipment and workstations are arranged sequentially in exact accordance with the order of operation required to build the product (Bellgran & Säfsten, 2010). Within the classical product-process matrix, the line layout is identified as the optimal choice for mass-producing highly standardised goods in very high volumes with minimal product variety. Material handling within this layout is typically managed by either a controlling conveyor without buffering or a floating conveyor with small intermediate buffers.

The continuous flow process represents the extreme end of the product-process spectrum (Holweg et al., 2018). Unlike discrete manufacturing, these systems process bulk materials or dimensional products measured in weight, volume, or length. In these environments, materials move continuously without stopping, guided through their transformative journey by highly specialised, tightly integrated, and perfectly synchronized equipment. The labour roles in this layout are typically monitoring and controlling the automated systems.

Manufacturing operations rarely fit precisely into a single process type. In practice, hybrid configurations are common, especially when different stages of production have conflicting requirements (Holweg et al., 2018). A typical example is a combination of batch processing upstream and either line or continuous flow processing downstream. The upstream stage is used to handle variation, where products differ in specification, sequence, or material. Further downstream, the focus shifts toward efficiency and consistency, with processes designed to operate at a more stable pace. This type of arrangement can be found across a range of industries, from furniture manufacturing to food processing, where components or semi-finished products are produced in batches and then assembled or finished in a more uniform flow.

What makes this configuration work is the presence of work-in-progress (WIP) inventory between the two stages (Holweg et al., 2018). This buffer is not simply a result of inefficiency, but a necessary part of the system. It absorbs differences in timing and pace between the process types and allows each stage to be planned more independently. The upstream batch process is often scheduled based on forecasts or stock replenishment needs, while the downstream stage can respond more directly to actual customer orders.

2.1.2 MTO as a Production Strategy

Manufacturing companies differ in the point at which a customer order begins to drive activities. Olhager (2003) defines the OPP as “the stage in the manufacturing value chain, where a particular product is linked to a specific customer order.” Production strategies such as make-to-stock (MTS), assemble-to-order (ATO), MTO, and engineer-to-order (ETO) correspond to different positions of the OPP along this value chain, see Figure 2.3 (Olhager, 2003). Activities downstream of the OPP are driven directly by customer orders, while upstream activities are primarily guided by forecasts and planning (Olhager, 2003). According to Olhager (2003), the position of the OPP is mainly determined by demand volume and volatility, as well as the relationship between customer delivery lead time and production lead time.

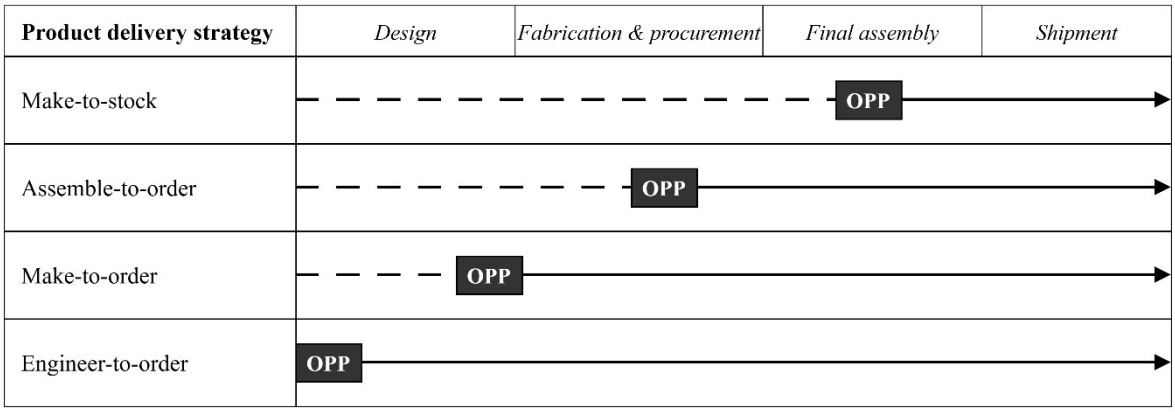


Figure 2.3
Different product delivery strategies relate to different OPP.

Comment. The dotted lines depict the production activities that are forecast-driven, whereas the straight lines depict customer-order-driven activities. Adapted from (Olhager, 2003).

In MTS production strategy, the OPP is located at the finished goods stage. Production is completed before any specific order is received, and products are stored in inventory until purchased by customers (Soman et al., 2004). In this context, the main competitive priority is maintaining a high fill rate, while planning challenges typically involve inventory management, lot sizing, and demand forecasting. In an MTO system, in contrast, no finished goods inventory is kept. Production starts only after a confirmed customer order is received, usually with an agreed due date (Soman et al., 2004). MTO systems allow a higher degree of product variety and customization, and production planning focuses more on order execution. Performance is commonly evaluated using measures such as response time, order delay, and due date adherence. In this setting, delivery lead time becomes more important than product availability.

Olhager (2003) also notes that competitive priorities differ between operations before and after the OPP. In forecast driven upstream activities, price tends to be the dominant concern, whereas delivery speed and flexibility are more important in downstream activities that respond directly to customer orders. As a result, production planning, performance measurement, and control practices need to be designed differently for each part of the value chain. This distinction also affects how material scrap can be measured and analysed. In the MTO portion of production, specifications, batch sizes, and routing sequences often vary from one order to another (Olhager, 2003; Soman et al., 2004). Process parameters such as cutting patterns and material quantities therefore change between jobs, and scrap levels may differ accordingly. These differences do not necessarily reflect process failures but can arise as a natural consequence of the MTO production strategy (Soman et al., 2004).

2.1.3 Level of Automation

Automation in manufacturing is not a question of whether a process is manual or automatic, but rather a matter of degree (Frohm et al., 2008). Frohm et al. (2008) define the level of automation (LoA) as "the allocation of physical and cognitive tasks between humans and technology, described as a continuum ranging from totally manual to totally automatic." In practice, most production systems can be placed somewhere between these two ends, with humans and machines sharing responsibilities within the same process.

Frohm et al. (2008) also suggest that level of automation (LoA) should not be treated as a single dimension. Instead, they distinguish between two separate aspects. The first is mechanical LoA, which relates to the extent to which physical tasks are carried out by equipment rather than by operators. The second is information LoA, which concerns cognitive activities such as monitoring, decision making and control, and whether these are performed by humans or automated systems. To describe these dimensions, Frohm et al. (2008) propose a seven-level reference scale, where level one represents fully manual operation and level seven corresponds to full automation. In real production environments, however, operations rarely sit at either end of the scale, and they are often positioned somewhere in between, with responsibility shared between operators and machines in both execution and control.

The level at which automation is implemented influences more than just technical performance. According to Bellgran and Säfsten (2010), it is one of the central dimensions used to characterise process technology in a production system, alongside factors such as scale and degree of coupling. Increasing mechanical automation can improve consistency and throughput, but it may also limit the system's flexibility when conditions change or when variation needs to be handled (Frohm et al., 2008). This becomes particularly relevant in MTO environments, where product specifications can differ from one order to another. Under such conditions, highly automated operations tend to perform well when tasks are stable and repetitive, while processes that must deal with variation often depend more on human involvement and judgement (Frohm et al., 2008).

2.1.4 Nesting in Sheet Metal Punching

In sheet metal manufacturing, the punching stage uses numerically controlled (NC) punch presses to produce blanks, which are intermediate parts cut from a sheet and subsequently formed through operations such as deburring and bending (Herrmann & Delalio, 2001). When a single production order is processed independently, it typically requires a pre-sheared sheet sized specifically for that order. Nesting changes this logic fundamentally. Herrmann and Delalio (2001) define nesting as the practice of combining multiple orders that require the same material type and sheet thickness, so that a punch press can complete all parts from those orders as one job. Instead of using pre-cut sheared sheets, nested orders run on larger standard unsheared sheets, which cost less per unit of weight because they do not require a prior shearing operation. The operator performs only one machine setup for the entire nest rather than one for each individual order, reducing both setup time and the number of sheets loaded.

From a materials perspective, nesting is fundamentally a 2-dimensional cutting and packing (CP) problem. Wäscher et al. (2007) provide a systematic typology of such problems, in which the objective is to assign small items (blanks or part geometries) to large objects (sheets) such that the geometric non-overlap condition is satisfied and a given objective function is optimised. In the cutting stock formulation relevant here, all part demands must be fulfilled and the objective is to minimise the total input value, which in practice means minimising the number of sheets consumed and the associated trim loss (Wäscher et al., 2007). When the objective function is expressed in terms of area, this is equivalent to waste minimisation, commonly referred to as trim-loss minimisation. The material that cannot be used for parts after punching, the trim loss, is embedded in the nesting layout (Herrmann & Delalio, 2001). It arises from the space between parts, the clamping area required at the sheet edges, and any residual area left after all parts have been placed. For a given set of orders and a given sheet width, the nesting software determines how many parts fit on each sheet and generates the corresponding numerically controlled (NC) program. The ratio of total part area to total sheet area consumed is therefore a direct output of the nesting decision, and constitutes the UTL metric. Finding the cost-optimal nesting decision

is an NP-complete problem (Herrmann & Delalio, 2001), which means commercial nesting software cannot guarantee optimal solutions and must instead rely on heuristics. As a result, the UTL recorded for any given production run reflects both the geometry of the parts being produced and the effectiveness of the nesting algorithm. Some level of trim loss is therefore always present, even under the best possible nesting conditions.

2.2 Lean Production

Lean production is a process-oriented approach aimed at creating maximum value for customers while minimising the use of resources and eliminating activities that do not contribute to value creation (Sörqvist & Bergendahl, 2021). Rather than being a set of isolated tools, lean represents a way of thinking about how processes are designed and driven, where the starting point is always the customer's definition of value.

A central concept in lean is the distinction between resource efficiency and flow efficiency (Sörqvist & Bergendahl, 2021). Resource efficiency refers to how well available resources such as machines and personnel are utilised, while flow efficiency refers to how much value is actually added to a unit relative to its total lead time through the process. These two dimensions can conflict with one another, and lean thinking prioritises flow efficiency, since a process can appear highly productive in terms of machine utilisation while still generating significant waste in the form of waiting, overproduction, or material loss (Sörqvist & Bergendahl, 2021). Achieving high flow efficiency requires that non-value-adding activities are identified and systematically reduced. In lean, such activities are referred to as waste, and their identification forms the analytical basis for improvement work. The types of waste relevant to manufacturing processes are described in the following section.

2.2.1 Waste Typology in Lean Production

Within lean production, the central analytical concept is waste, referred to in Japanese as muda. It is generally understood as any activity that consumes resources without creating value for the customer (Ohno, 1988). Ohno (1988) identified seven categories of waste seen in Table 2.1. Among these, overproduction is often highlighted as most problematic, since it tends to both generate and conceal other forms of waste, including inventory build-up and quality related issues (Liker, 2021).

Table 2.1
The seven types of waste in lean production

Waste Type	**Definition**
Overproduction	Producing earlier/faster/more than is needed
Waiting	Idle time for people, machines, or material
Unnecessary Transport	Unnecessary movement of material/products
Overprocessing	Doing more work or using higher spec than required
Excess inventory	Excess WIP or finished goods beyond what is needed right now
Unnecessary movement	Unnecessary movement of people
Defects	Products not meeting requirements, leading to scrap or rework

From the customer's point of view, activities within a process can be divided into 3 broad groups: value-adding activities, non-value-adding activities that are still necessary under current conditions, and activities that neither add value nor serve a necessary function and should therefore be eliminated (Womack & Jones, 2003). Hines and Rich (1997) further developed this classification by linking each of the seven waste types to specific diagnostic tools. For instance, waiting is described as periods where materials or products are idle, and excess inventory is linked to longer lead times as well as the tendency to hide underlying process issues.

2.3 Production Flow and Process Mapping

Production flow describes how materials, information, and products move through a sequence of activities from initial input to finished output (Bellgran & Säfsten, 2010). To analyse such flows, processes are often represented as a series of discrete operations, inspections, transports, and waiting points, making their structure and interdependencies visible. Process mapping refers to this systematic visualisation of activities and interfaces, providing a basis for understanding how work actually progresses through a production system (Rother et al., 2003).

Within lean production, the concept of value provides a starting point for analysing flows. Womack and Jones (2003) define value as a capability provided to the customer at the right time and at an appropriate price, emphasising that it must be specified by the end customer rather than by internal functions. From this perspective, the value stream becomes the primary unit of analysis, representing the full set of activities required to transform raw material into a product the customer is willing to pay for (Womack & Jones, 2003). When the value stream is mapped, activities can be grouped into value-adding operations, non-value-adding but currently necessary activities (often referred to as Type 1 waste), and purely non-value-adding activities that can be eliminated directly (Type 2 waste). Distinguishing between these categories is essential for prioritising improvement efforts, since Type 2 waste can be removed immediately whereas Type 1 typically requires changes to the underlying system.

Several authors argue that analysing processes from the customer's viewpoint is crucial for separating value-adding work from activities that only consume time and resources. Liker (2021) highlights that a customer-oriented view frequently reveals more waste than internal efficiency assessments, since it challenges assumptions about what is truly needed to deliver the product. This orientation shifts attention from what is technically required by the producer to what is actually required to satisfy customer needs (Liker, 2021).

Building on this perspective, Hines and Rich (1997) introduce a set of structured tools for value-stream analysis, among which process activity mapping is particularly relevant. Process activity mapping records each operation, transport, inspection, and delay in a production flow, typically including information on time and inventory (Hines & Rich, 1997). By quantifying these elements, the method makes non-value-adding time and waiting visible, rather than relying on qualitative estimates. Their case studies show that a large share of total lead time can consist of material waiting in queues or storage even when actual processing time is relatively small, illustrating how inventories and delays can distort perceptions of flow efficiency.

In manufacturing contexts concerned with waste reduction, process mapping therefore serves not only as documentation but as an analytical technique for identifying where losses originate (Rother et al., 2003). By explicitly capturing material movements, process interfaces, and buffer locations, mapping enables practitioners to locate stages where material is transformed into scrap, where rework or handling

occurs, and where inventories accumulate (Hines & Rich, 1997). As such, process mapping is widely used as a first step in waste reduction and material-efficiency initiatives, providing the foundation needed to design process-level performance measures and to target subsequent improvement efforts (Hines & Rich, 1997).

2.4 Material Scrap

In manufacturing processes, not all consumed material ends up in the final product, as part of it becomes residual or waste material (Milford et al., 2011). Milford et al. (2011) estimate that at a global scale, around 26% of all liquid steel never reaches a finished product, instead being lost as process scrap during stages such as fabrication and forming. While this figure relates to primary metal industries, it reflects a broader pattern that can also be observed in downstream manufacturing. Material scrap arises at multiple stages, and when losses at every stage are considered together, they lead to a significant reduction in overall system yield.

Material efficiency is defined as the ratio between useful product output and the raw material input required to produce it (Shahbazi et al., 2016). Improving material efficiency therefore means reducing the volume of residual material generated per unit of output. Shahbazi et al. (2016) identify material efficiency improvement as a key strategy for reducing industrial waste volumes, limiting resource extraction, and lowering energy demand and emissions. Despite this importance, material efficiency in manufacturing remains under-researched relative to energy efficiency, and companies face significant barriers to measuring and improving it in practice (Shahbazi et al., 2016).

2.4.1 Types of Scrap

Material scrap in manufacturing can be divided into two categories based on its origin. The first type is normal spoilage, defined as waste that is built into the production process and occurs even under efficient operating conditions (Datar & Rajan, 2017). Because this type of loss is built into the process design, its costs are absorbed into the cost of good units rather than treated as a performance failure. Juran (1986) describes the same phenomenon as chronic waste, noting that it is "planned into the process, in the sense that the planning process failed to plan it out." Reducing it requires intervention at the design level, since operators working within the existing process cannot eliminate it through better execution alone. As Datar & Rajan (2017) put it, managers make a conscious decision about production rate "with the understanding that, at the chosen rate, a certain level of spoilage is unavoidable."

The second category is abnormal spoilage, defined as waste that falls outside what the production process is expected to generate under normal, efficient conditions (Datar & Rajan, 2017). Unlike normal spoilage, this type of loss is considered avoidable and controllable. In practice, it can be reduced, or in some cases eliminated, by identifying and addressing underlying causes such as machine breakdowns, incorrect settings, and other process deviations. Juran (1986) describes these as sporadic problems, meaning sudden deviations from normal process conditions that require immediate action to restore operations. At the same time, Juran (1986) highlights an important imbalance in how organisations approach improvement. Efforts are often concentrated on these visible and immediate disruptions, while more persistent, chronic sources of waste receive less attention, despite often representing a larger share of the total cost. From this perspective, reducing abnormal scrap can be seen as part of day-to-day operations management, whereas addressing planned or chronic scrap requires interventions at the design or engineering level.

2.4.2 Challenges in Measuring Scrap

Despite its importance, measuring scrap at the process-level is associated with several practical challenges. Shahbazi et al. (2016) group the most influential barriers to improving material efficiency into four areas: budget constraints, information-related issues, management priorities, and employee-related factors. Among these, the information barrier is the most relevant in this context. It includes limited knowledge sharing, poor visualisation of material flow, and insufficient data availability. As a result, companies may be aware of their total waste volumes, but still lack the ability to link material losses to specific operations (Shahbazi et al., 2016). This information gap can also affect how employees perceive scrap distribution within the total volume. Tversky and Kahneman (1974) describe the availability heuristic as a cognitive tendency where perceived frequency of events is shaped by how easily instances come to mind. In production environments where process-level data is absent, subjective assessment may therefore overrepresent operations that generate very visible or physically handled scrap, rather than reflecting the actual distribution.

Controlling process scrap requires first identifying its sources before any meaningful quantification can take place (Santos et al., 2025). However, approaches to measuring and categorising quality related costs vary across companies and industries, with no single standard methodology in widespread use (Schiffauerova & Thomson, 2006). This means that measurement approaches need to be adapted to the specific process and organisational context in which they are applied. The challenge becomes more pronounced in MTO environments, where variation in product specification and batch sizes makes it harder to define a consistent measurement baseline. It also complicates the distinction between scrap caused by process and scrap that is inherent to the product design.

2.5 Cost of Quality

The financial consequences of material scrap in manufacturing can be understood within the broader framework of the cost of quality (CoQ). CoQ is generally understood as the sum of conformance and non-conformance costs, where the cost of conformance covers prevention and appraisal activities and the cost of non-conformance covers the costs caused by product and service failures (Schiffauerova & Thomson, 2006). The most widely studied and implemented model for structuring these costs is the prevention-appraisal-failure (PAF) model.

Within the PAF framework, costs are divided into four categories. Prevention costs are defined as the costs of all activities specifically designed to prevent poor quality in products and services. Appraisal costs are the costs associated with measuring, evaluating or auditing products or services to assure conformance to quality standards and performance requirements. Internal failure costs are the costs resulting from products or services not conforming to requirements prior to delivery to the customer, and external failure costs are those arising after delivery (Plewa et al., 2016). Scrap falls explicitly under internal failure costs and is defined by De Feo (2017) as the labour, material and overhead on defective or non-conforming product that cannot be repaired, reused or resold.

The relative weight of internal failure costs is significant. De Feo (2017) notes that in documented case studies, internal failure costs have been found to dominate total CoQ, in one example accounting for 79.4 % of the combined quality cost figure. This distribution carries a direct implication for improvement prioritisation: reductions in internal failure costs, including scrap, represent the largest financial opportunity. At a broader level, De Feo (2017) reports that the cost of poor quality in

manufacturing organisations commonly amounts to 15 to 30 % of all costs, a figure large enough to exceed annual profit in many cases.

Despite this potential significance, quality costs are frequently underreported or hidden within general accounting categories. Schiffauerova and Thomson (2006) note that companies typically measure visible and quantifiable costs such as scrap and warranty claims but ignore costs that are more difficult to trace, meaning a high proportion of quality costs remain hidden. De Feo (2017) specifically identifies that scrap may go unreported because employees fear reprisals or because the cost is absorbed into general ledger accounts without being explicitly identified as scrap.

Empirical evidence from manufacturing companies supports the practical relevance of the PAF model. Plewa et al. (2016), in a regression analysis of 286 observations from a homogeneous sample of German manufacturing companies, found that total CoQ averaged 2.7 % of net sales, with failure costs accounting for 47.8 % of total CoQ on average. Their study confirmed that lower overall quality levels are significantly associated with higher failure costs, and that a higher share of failure cost in total CoQ is associated with lower overall quality performance. These findings reinforce the argument that reducing internal failure costs, such as process-generated scrap, produces measurable quality and financial benefits.

The CoQ framework is relevant to this study not as a measurement tool applied directly, but as a theoretical basis for understanding why process-level scrap visibility matters beyond operational performance. Without measurement, the costs associated with scrap cannot be identified, quantified or prioritised for improvement. Establishing measurement capability at the process-level is therefore both a precondition for DMAIC-based improvement work, and the necessary first step in making hidden internal failure costs financially transparent.

2.6 Performance Measurement

Performance measurement can be defined as the process of quantifying the efficiency and effectiveness of actions, while a performance measure is a metric used to quantify that efficiency or effectiveness (Neely et al., 1995). A performance measurement system is the set of such metrics used together to give a structured view of how well an organisation is achieving its objectives. A fundamental distinction in the design of performance measurement systems is the difference between lagging and leading indicators (Neely et al., 2000; Parmenter, 2020). Lagging indicators, also referred to as result indicators, measure outcomes after a process has occurred and are relatively straightforward to collect but offer limited opportunity to intervene in real time (Parmenter, 2020). Leading indicators, by contrast, measure process conditions or behaviours that are predictors of future results and are harder to capture but enable earlier corrective action.

A second challenge relevant to this study is the distinction between aggregate and process-level measurement. Aggregate measures combine outputs from multiple process stages into a single metric, which simplifies reporting but conceals the stage-specific origin of losses (Neely et al., 1995). Process-level measurement, by contrast, attributes performance outcomes to the individual step at which they arise, enabling more targeted improvement decisions. In scrap management, an aggregate scrap figure for a production period tells a manager how much material was lost but not which process step generated it or why. Decomposing that aggregate into process-stage contributions is precisely the objective of the measurement model developed in this thesis, and it is this decomposition that makes the findings actionable rather than descriptive.

2.7 Sources of Variation

All manufacturing processes produce some degree of variation. Even under stable conditions, the output will differ from one unit to the next (Montgomery, 2008). Materials, machines, operators, methods, and measurement systems, as well as environmental factors such as temperature and humidity, are identified as main sources of this variation. Each of these contributes in small ways to the overall variability observed in product characteristics, whether in dimensions, weight, or surface finish. As a result, no two units produced by the same process are ever completely identical.

To understand and manage variation, Montgomery (2008) distinguishes between two types of variation. The first consists of chance causes, which are the many small and unavoidable sources of variation that are always present in a process. When only these are acting, the process is considered to be in statistical control, meaning that its behaviour is stable and predictable within known limits. The second type is assignable causes (Montgomery, 2008). These are specific factors that are not part of normal operation and that lead to deviations beyond the expected level of variation. They are usually larger in effect and can typically be identified and corrected. Examples include tool wear beyond tolerance, material that does not meet specifications, or incorrect operator actions. When assignable causes are present, the process is no longer stable and considered to be an out-of-control process. The distinction between these two types of variation is important because it determines how the process should be managed (Montgomery, 2008). If only chance causes are present, the process is already performing at the level allowed by its current design. Reducing variation further then requires changes to the process itself, for example through improved equipment, better materials, or revised process design. When assignable causes occur, the situation is different. These deviations should be investigated and corrected at the operational level, since removing them restores the process to its normal, predictable state.

From a cost accounting perspective, this framework parallels the distinction between normal and abnormal spoilage introduced in Section 2.4.1 (Datar & Rajan, 2017; Montgomery, 2008). Normal spoilage corresponds to scrap arising from chance-cause variation that is inherent in the current process design, whereas abnormal spoilage reflects assignable-cause variation that should be detected, investigated, and reduced through targeted corrective actions. Framing scrap in this way links statistical process control to the economic treatment of spoilage and provides a common language for later analysis of material losses.

2.8 Continuous Improvement

Process improvement is a fundamental activity in operations management, alongside planning and resource management (Holweg et al., 2018). In most industrial settings, operations exist in “brownfield” environments where processes already exist and must be continuously adjusted rather than designed from scratch. Holweg et al. (2018) explain this through the concept of operational “entropy”, which is the tendency for unmanaged processes to deteriorate over time as non-standard workarounds accumulate, variation increases and performance declines. Structured improvement is therefore not only

a response to conform to market pressure but also seen as a tool to counter the process deterioration, as shown in Figure 2.4.

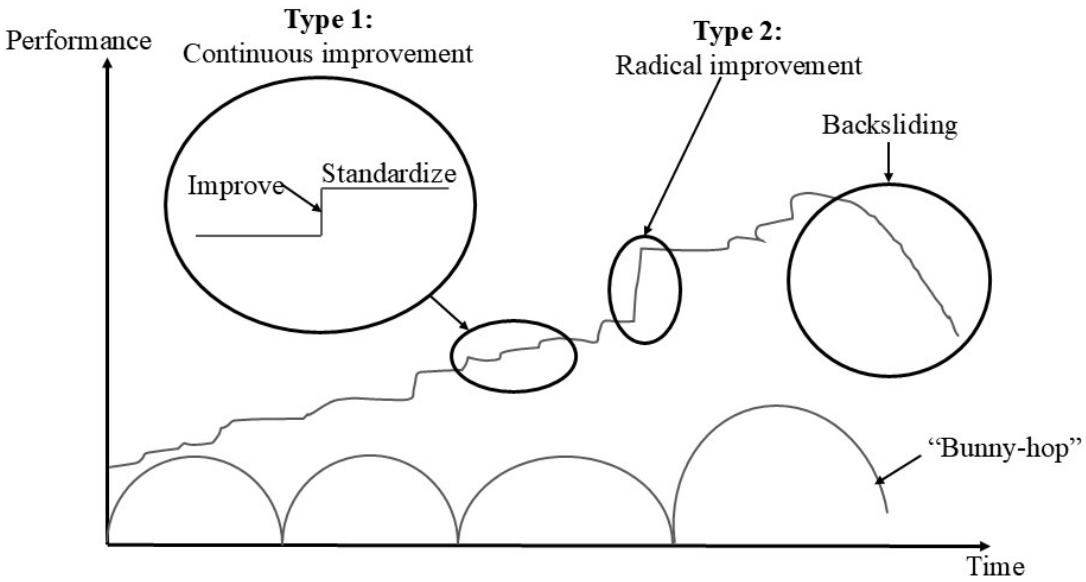


Figure 2.4

Process improvement entropy

Comment. Adapted from (Holweg et al., 2018)

Holweg et al. (2018) distinguish between two basic forms of process improvement. Continuous improvement (kaizen) refers to ongoing incremental changes driven by workers within the existing process. Breakthrough or radical improvement (kaikaku or rapid improvement events), involves more radical changes such as modifications to the system or layout of the production, and becomes necessary when incremental efforts reach a performance ceiling. Neither approach is sufficient alone, radical improvements are unsustainable without the continuous improvement structure to maintain the gains, while incremental improvement alone may fail to address inherent constraints of a production system. A well-established application of kaizen logic to batch-based manufacturing processes is the reduction of changeover frequency and duration. Shingo (1985) formalised this through the concept of Single-Minute Exchange of Die (SMED), which distinguishes between internal setup activities that can only be performed while a machine is stopped, and external activities that can be prepared in advance while the machine is still running. By converting internal activities to external ones and simplifying those that remain internal, changeover time can be substantially reduced without capital investment. The underlying principle extends beyond the physical setup operation itself: when changeover costs are reduced, smaller and more frequent productions become economically viable, which in turn reduces the need to batch large quantities of the same material at once. In MTO sheet metal manufacturing, where scrap generation varies between orders and across process stages, structured improvement requires the identification of where losses occur and establishing the measurement capabilities to track them over time. Without this foundation, improvement efforts risk targeting the wrong operations or failing to sustain gains once achieved.

2.8.1 Six Sigma as a Data-Driven Improvement Philosophy

Six Sigma is a methodology for process improvement which focuses on systematic reduction of variation and defects (Linderman et al., 2003). Linderman et al. (2003) define Six Sigma as an organised and systematic method for process improvement that relies on statistical methods and the scientific method to make dramatic reductions in customer defined defect rates. Holweg et al. (2018) describe Six Sigma as evolving from statistical process control into a broader management philosophy, focusing on seeing variation as the main culprit to low process performance. Basu (2023) further describes Six Sigma as a data-driven methodology that relies on measurable processes and emphasises the method's financial impact when applied to processes.

The relevance of Six Sigma in a manufacturing context stems from the tendency of unmanaged processes to accumulate variation over time. As described in the previous section, process entropy causes production systems to deteriorate without continuous intervention. Six Sigma addresses this by providing a structured improvement cycle that forces decisions to be grounded in data rather than intuition, making it particularly well suited to contexts where scrap generation is variable, order-dependent and difficult to attribute without systematic measurements.

2.8.2 DMAIC

DMAIC is the main problem-solving framework within Six Sigma and provides a structured, sequential approach to process improvement (Basu, 2011). The approach is designed for application to already established processes where a performance gap has been identified but the root causes are unknown. Unlike less structured improvement approaches, DMAIC requires that each phase is completed and its outputs verified before the next phase can begin, ensuring that improvement decisions are grounded in data rather than assumption (Basu, 2011). The five sequential phases are: Define, Measure, Analyse, Improve and Control. The Define phase establishes the foundation of the improvement project. Its purpose is to clearly articulate the problem, set the project scope and identify the performance indicator that will be used to track improvement (Sörqvist & Höglund, 2017). Several tools can be applied during this phase, including process flowcharts, value stream mapping (VSM) and project charters. While VSM can provide a detailed visualisation of the process flows across a value stream, it is primarily oriented to address flow inefficiencies rather than establishing project scope. A Supplier, Input, Process, Output, Customer (SIPOC) diagram is therefore commonly preferred as the primary tool for establishing the scope of the project, since it provides a good overview of the process without requiring the data collection that VSM demands at an early stage. The Define phase is finished when the problem is clearly stated, the relevant process boundaries are agreed upon, and a measurable target has been determined.

The Measure phase of the project establishes the quantitative baseline of the current process performance (Sörqvist & Höglund, 2017). This involves determining what data is needed, identifying where and how it can be collected, and validating that the measurement system is reliable. Available approaches range from automated data collection from existing digital production systems like Enterprise Resource Planning (ERP) platforms, to manual data collection methods like direct measurements or operator logging. Automated sources generally provide increased data reliability due to higher level of traceability and lower risk of measurement error, while manual methods may be necessary where automated data does not exist (Basu, 2011). Data collection should be targeted at the specific process steps identified in the Define stage rather than at the aggregate system level, to enable comparison between the process steps. The output of the Measure phase is a documented baseline against which future improvements can be evaluated, and a confirmation of the gap between the baseline and target.

The Analyse phase uses the baseline data as input and applies analytical techniques to identify and verify the root causes of the performance gap (Sörqvist & Höglund, 2017). To identify root causes, hypotheses are systematically generated and tested to determine what drives the observed variation or performance loss. Basu (2011) describes several tools applicable to this phase, including regression analysis for identifying relationships between process variables, five whys technique to establish possible root causes from system symptoms, and inter-relationship diagrams for mapping connections between multiple factors. The selection of applied tools depends on the nature of the problem and what data is available. The phase is finished when the most influential root causes have been identified and verified using data, providing a solid basis for the next phase.

The Improve phase bridges between the findings of the Analyse phase and targeted countermeasures (Sörqvist & Höglund, 2017). Potential solutions are generated through structured measures such as brainstorming or benchmarking against similar processes, and evaluated against performance indicators such as expected impact, feasibility and implementation cost. An action plan is developed and where possible, potential solutions are tested before full implementation to reduce the risk of unintended consequences (Basu, 2011). The output of the Improve stage is a set of implemented changes that address the verified root causes and move process performance closer to the target.

The Control phase ensures that improvements are sustained after the project concludes (Sörqvist & Höglund, 2017). This involves updating the standard procedures and establishing monitoring systems to track the chosen performance indicator over time and defining response plans in case the performance indicator regresses. Tools in this phase include Gantt charts for tracking the progress of implementation activities, activity network diagrams for mapping task dependencies and identifying the critical paths, radar charts for visualising performance across several indicators, and the Plan-Do-Check-Act (PDCA) cycle for structuring ongoing monitoring and incremental adjustment. Together these tools can support the transition between active improvement work to stable, continuous improvement. Without a structured Control phase, processes tend to revert to their previous state.

2.8.3 Measurement System Design and Validation

A fundamental precondition for the Measure phase of DMAIC is that the data collected must accurately reflect the process being studied. A measurement system can be understood as the combination of the instrument, method, operator, and the conditions under which data is collected (Montgomery, 2008). Each of these elements can introduce variability that is unrelated to the process being measured, meaning that the measurement system itself is a potential source of variation that must be evaluated before process-level conclusions can be drawn.

For a measurement system to be considered reliable, two properties must hold (Montgomery, 2008). The first is repeatability, which refers to the degree to which the same measurement method produces consistent results when applied to the same object or quantity under the same conditions. The second is reproducibility, which refers to the consistency of results across different operators, time periods, or measurement conditions. Together, these two properties determine whether the data produced by a measurement system reflects actual process behaviour or whether it is obscured by variability introduced through how the measurement is conducted. When measurement variation is large relative to the process variation being studied, the data becomes an unreliable basis for identifying root causes or evaluating whether an improvement has had any effect.

Evaluating the extent to which a measurement system satisfies these properties is therefore a necessary step before any process data can be used as the basis for analysis. This process, known as Measurement

System Analysis (MSA), becomes particularly challenging in production environments where not all process stages generate digital records automatically. Frohm et al. (2008) distinguish between the physical and cognitive-related dimensions of automation as two independent continuums. At the lowest cognitive LoA, process data collection relies entirely on the operator's own experience and judgement, with no automated capture or storage of information. As cognitive automation increases, so does the degree to which information is systematically collected, tracked and used to control the process. In a production environment where certain stages operate at a low cognitive level of automation, the measurement system cannot simply be inherited from existing infrastructure but must be actively designed to compensate for the absence of automatic data generation. Shahbazi et al. (2016) provide empirical support for this by identifying lack of information and measurement methodology as among the most influential barriers to process-level improvement in manufacturing.

The standard method for evaluating a measurement system is a Gauge Repeatability and Reproducibility study (Gauge R&R), in which the contributions of instrument, operator, and conditions to total measurement variation are quantified through a controlled experiment (Montgomery, 2008). When such a study is not feasible due to the absence of a defined instrument or stable measurement conditions, the use of multiple independent data sources that overlap at shared measurement points offers an alternative basis for assessing consistency. Where independent sources produce converging results, this supports the construct validity of the overall measurement approach, while divergence signals either a gap in coverage or a source-specific bias that requires further investigation (Yin, 2018). In a hybrid MTO production environment where different stages require fundamentally different measurement approaches, such convergence across sources becomes the primary available basis for establishing confidence in the combined measurement design before data from that system can be used as the basis for analysis.

2.8.4 Root Cause Analysis

Root cause analysis is a structured approach to identifying the true cause of a problem and the actions required to eliminate it (Andersen & Fagerhaug, 2006). It is recognised as a core component of continuous improvement work and forms an integral part of any structured problem-solving process. A central distinction in root cause analysis is the recognition that problems rarely have a single, immediately visible cause. What is first observed is typically a symptom rather than a cause, and behind any visible problem there is a chain of cause-and-effect relationships linking first-level causes to deeper, higher-level causes. The root cause sits at the top of this chain and represents the point at which, if eliminated, the problem would be prevented from recurring. Addressing only symptoms or first-level causes may temporarily reduce the visible problem, but without reaching the root cause the underlying condition will re-emerge in some form.

Root cause analysis is not a single method but a collective term for a range of tools and techniques applied across the different phases of structured problem solving, spanning problem understanding, cause brainstorming, data collection, data analysis, root cause identification, root cause elimination, and solution implementation (Andersen & Fagerhaug, 2006). Within the root cause identification phase, the cause-and-effect diagram combines brainstorming with structured visual analysis to generate and systematically evaluate possible causes of a problem. The five whys technique approaches causality iteratively, following each identified cause by asking why it occurred and repeating this process until no deeper cause can be found. The matrix diagram maps relationships between multiple potential causes and problem characteristics to assess which causes carry the greatest overall contribution to the observed

problem. The selection between these tools depends on the nature of the problem and the analytical context, and in practice a root cause analysis often requires several tools to be applied in sequence or iterated before a verified root cause is reached.

A practical challenge in conducting root cause analysis is that the process is highly dependent on the quality and completeness of the data and process knowledge available to the team (Andersen & Fagerhaug, 2006). Where process-level data is absent or fragmented, identifying whether a cause is a genuine root cause or merely a first-level manifestation of a deeper problem becomes difficult. In such contexts, multiple iterations through the cause identification tools may be needed before the analysis converges on a cause that is both operationally traceable and actionable.

3 Methodology

This chapter describes the methodological approach taken to address the three research questions. Since the research questions span both the definition and quantification of scrap and the identification of sources of variation, a combination of qualitative and quantitative methods was required. To ensure coherence between the study's purpose and its methodological approach, the research design and data collection methods were aligned with the three research questions. Each research question required a different type of data and a specific analytical approach, which informed the selection of methods used within the study. RQ1, which concerns how scrap and UTL can be defined and measured at process-level, was addressed through direct observations, ERP data extraction and manual scrap bin weighing. Observations established the process boundaries and identified scrap-generating stages, which defined the scope of the measurement system. ERP extraction and manual weighing then provided complementary coverage of scrap volumes across those stages, with the combined design forming the measurement system that RQ1 seeks to establish. RQ2, which concerns the challenges and sources of variation affecting measurement and control, was addressed through semi-structured interviews and the quantitative data collected in the Measure phase. The interviews provided qualitative accounts of perceived causes, while the ERP and offcut data enabled these accounts to be tested against measured variation patterns. RQ3, which concerns how process-level measurement can support the identification and proposal of scrap reduction measures, was addressed analytically through the Improve phase, drawing on the root causes identified in RQ2, and the baseline established in RQ1 to develop and assess the feasibility of targeted improvement proposals.

3.1 Research Approach

The study followed a deductive approach, in which the established theoretical frameworks guided the structure of the investigation rather than emerging from the empirical findings. The DMAIC framework was adopted as the overarching structure, with each phase defining the scope and sequence of data collection and analysis. Theoretical concepts drawn from the literature on scrap measurement, MTO production, and process variation informed the design of the data collection instruments and the analytical categories applied in the study. Empirical findings were subsequently used to test and apply these frameworks within the specific production context, rather than generate new theory (Bryman, 2016). This approach is appropriate given that the study operates within an established methodological framework and addresses a defined practical problem, where the aim is to apply and evaluate existing knowledge rather than to discover new theoretical constructs (Creswell, 2009).

3.2 Research Design

The selection of a research design provides a structured approach to conducting a study, ensuring that the formulated research questions, purpose and methodological approach align (Yin, 2018). Without a clearly defined research design, the study risks producing data collection and analysis that is disconnected from the study's purpose and aim. A single-site case study was therefore adopted, following the framework described by Yin (2018). A case study research design was chosen because it enables investigation of phenomena within its real-world context, which is particularly well suited to understanding complex production behaviours such as scrap generation where contextual factors are

difficult to separate from the problem itself (Creswell, 2009). The control phase is excluded from this study, as discussed in section 1.5.

To enable a fully developed analysis of the research problem, a concurrent mixed-method approach was used, combining qualitative and quantitative data collection throughout the study. Qualitative data, gathered through observations and interviews, was used to map the production flow, identify possible scrap sources and understand process-level challenges. Quantitative data, gathered through ERP extraction and manual scrap measurements, was used to determine scrap volumes, compare process stages and identify variation patterns. The combination of both data types allowed for triangulation, where findings from one source could be verified or challenged by another, reinforcing the overall reliability of the study (Creswell, 2009). The concurrent mixed-method approach carries both advantages and limitations. Its principal advantage is that qualitative and quantitative data can compensate for each other's weaknesses: qualitative data captures contextual factors and perceived causes that cannot be quantified, while the quantitative data provides the empirical rigour needed to test and validate those accounts against measured patterns. A limitation of this approach is that integrating two data types increases the complexity of both data collection and analysis, and the risk that the two streams produce contradictory findings must be managed through cross-validation rather than assumed away (Creswell, 2009).

3.3 Case Company

The case company for this study is a manufacturer of air handling units (AHU) and cooling baffles with a production facility located in southern Sweden. The facility was selected for this study through prior contact established during a summer internship at the company's quality department, which provided access to the production environment and insight into the development of improvement initiatives, which this study is one of. The studied production line is dedicated to producing insulated panel wall subassemblies, which form the core insulating component of the walls of the AHU produced at the facility. The line operates as a hybrid between batch-based and line-based production. The upstream stages: decoiling, coil levelling and cutting, operate on a batch basis whilst the downstream stages: automated punching, bending, insulation assembly, robot welding and sealing strip installation, follow a line-based flow. All panels are produced to customer order, meaning that dimensions and specifications vary between orders, introducing variability in UTL.

This production context is directly relevant to the purpose of this study. The combination of MTO production, hybrid process structure and the absence of process-level scrap measurements create precisely the conditions under which the research questions were formulated. The variability introduced by order-specific specifications and the configuration differences between the batch and line segments make process-level measurements both necessary and non-trivial, providing a rich and representative context for applying DMAIC as a structured improvement framework. Data was collected through observations, semi-structured interviews, ERP data extraction and manual scrap bin measurements conducted at the facility, as described in the following sections.

3.4 Data Collection

Data was collected through five methods: structured observations, semi-structured interviews, ERP data extraction, coil offcut measurements, and manual scrap bin measurements, and the connection between

these methods and the three research questions is illustrated in Figure 3.1. All data was collected as primary data, either directly from the production environment or from the case company's internal systems. Observations and interviews were conducted during the Define phase to establish an understanding of production flow and identify potential sources of scrap, while ERP data extraction, coil offcut measurements, and manual scrap bin measurements were conducted during the Measure phase to quantify scrap generation across individual process stages. The combination of qualitative and quantitative data sources enabled triangulation, where the findings from one source could be verified or challenged by another, strengthening the overall reliability of the study (Yin, 2018). The collection method, volume of data and collection phase are summarised in Table 3.1.

Table 3.1

Summary of data collection methods, sample sizes, and associated DMAIC phases

Method	Volume	DMAIC phase
Observations	3 sessions, ~45 min each	Define
Semi-structured interviews	4 participants, ~10 to 30 min each	Define
ERP data extraction	134,774 records (28,214 complete)	Measure
Scrap bin measurements	122 measurements	Measure
Coil offcut measurements	14 measurements, 4 production days	Measure

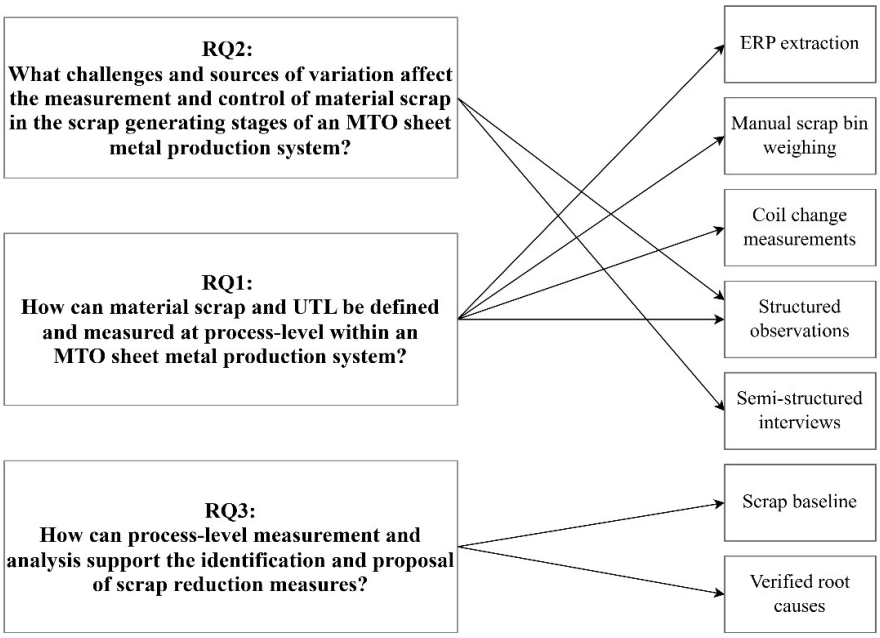


Figure 3.1

Connection between Methodology and RQs

3.4.1 Observations

Direct observations were conducted at the case company's manufacturing facility during the Define phase to build a deeper understanding of how the production operates and where scrap arises on a process-level. Observations serve as a systematic method for collecting firsthand data on real phenomena, particularly useful in complex production environments where process behaviour cannot be fully captured through documents or interviews alone (Yin, 2018).

Each observation session targeted a specific segment of the production system. The first observation session covered the upstream batch segment, including coil handling, coil levelling and cutting. The second session covered the line-based segment including the selection table, automated punching, bending and insulation assembly. The third session covered the final segment of the observed production system, including blank removal station and automated storage. This segmentation was deliberate as the three sessions correspond to the three different phases of the production system: pre-fabrication, panel component production and final storage and assembly. Three observations were considered sufficient to establish a comprehensive overview of the production system and where potential scrap sources were.

Each observation was guided by a structured protocol divided into four categories: process properties, flow properties, scrap generation, measurement feasibility. For each observation session, the protocol captured which process steps were visible, which could be seen generating scrap, the automation level of each step, whether idle time existed between steps, the type and frequency of scrap produced, whether machine settings were static or variable and how scrap was currently being measured or how it could be measured. Following each session, the findings were discussed between the authors to build understanding and identify areas that required follow-up. The observations provided the foundational understanding of the production flow and determined the scope of data collection for the Measure phase.

3.4.2 Semi-Structured Interviews

To complement the observations, semi-structured interviews were conducted with four individuals in roles related to the production process: one team leader, two quality engineers and one operator. The respondents were chosen through purposive sampling, ensuring that the selected participants possessed relevant knowledge and experience of the production process from different organisational perspectives (Yin, 2018). Two quality engineers with differing specialities were chosen as interviewees, one with knowledge about production quality, and one with specialised knowledge about the processes at the case company. The interviews were conducted during the Define phase to capture process-specific knowledge and challenges that were not visible through observations alone.

The semi-structured format was chosen since it combines a prepared theme framework with the flexibility to capture both anticipated and unanticipated insights, making it well suited for discovery-oriented data collection in complex environments (Magaldi & Berler, 2020). This approach was considered more appropriate than fully structured interviews, which could limit the depth of the responses, or unstructured interviews, which would severely reduce the comparability between the respondents. Each interview was conducted as a discrete meeting and was audio recorded for accurate transcription, with informed consent obtained from all participants prior to recording. Interview duration ranged from approximately ten to thirty minutes, depending on the depth of the questions relevant for each respondent's role and the respondent's communication style.

3.4.3 Quantitative Data

Quantitative data was collected during the Measure phase from three sources: the company's ERP system, manual scrap bin measurements, and coil offcut measurements. Together these sources provided scrap data across the production stages, though with differing levels of traceability. ERP data from the punching stage could be traced to individual order and article types, while manual scrap bin measurements from the batch-based segment captured overall scrap volume per stage without order-level traceability.

The primary source of quantitative data was extracted from the company's ERP system via direct SQL queries, where the automatic nesting program has historically logged metrics such as sheet dimensions, production dates and material utilisation on a per-article basis. The extracted parameters were article number and variant, production date, coil identifying number, coil width, material surface density, cut sheet dimensions and punch UTL. The extraction produced a dataset comprising production records with production dates spanning from April 2025 to March 2026. Not all records contained complete UTL data, corresponding to the periods during which process-level measurement was actively conducted on the production line. Outside these periods, data logging was either inactive or incomplete and was therefore excluded from the analysis. The extracted data was imported into Excel for compilation and analysis. The selected performance indicator for this study is UTL percentage, which is directly available from the ERP system. By combining the UTL data with sheet area and material surface density, scrap weight per article could be calculated where needed to provide context for the data.

For production stages that were not covered by the ERP system, scrap data was collected manually by weighing scrap bins before emptying. Logistics personnel responsible for transporting scrap were provided with a data collection sheet and instructed to record the date, weight, where the scrap was accumulated and of which material the scrap consisted of each time a bin was emptied. Data collection varied by stage: scrap bin measurements at the coil cutting stage were available for February 2026 only, while measurements from the punching stage and blank removal station covered the full collection period from February to March 2026. The collected sheets were compiled in Excel. Although this data cannot be traced to individual orders, it still provides a comparable overview of scrap production that supplements the ERP data. To quantify scrap generation specifically at the coil cutting stage, coil offcut measurements were captured. Each time a coil was changed, the length of material trimmed and discarded at the start of the new coil was measured and recorded by the authors on a structured data collection form. The form captured operator number, coil number, trimmed length and the date and time of measurement. The collected measurements provide the empirical basis for estimating coil cutting scrap.

3.5 Data Analysis

This section describes the analytical approach applied to the qualitative and quantitative data collected during the study. The two data types required different analytical methods, reflecting the different nature of the data and its role in addressing the research questions. Qualitative data was analysed thematically to identify patterns across observations and interviews, while quantitative data was processed statistically to establish a process-level scrap baseline and identify variation patterns.

3.5.1 Qualitative Analysis

The qualitative data collected through observations and semi-structured interviews was analysed using thematic analysis, following the 6-step process outlined by Braun and Clarke (2006). Thematic analysis was chosen since it provides a systematic approach for identifying, analysing and interpreting patterns within qualitative data, whilst affording sufficient flexibility to suit explanatory data collection within complex production environments.

The analysis proceeded through six steps. The transcripts were first re-read to establish a familiarity with the material before any coding began. While the coding process remained data-driven, the theoretical framework provided a background understanding that informed how emerging themes were interpreted and named. Relevant passages were then identified and assigned initial codes organised into candidate themes, which were reviewed and refined iteratively to ensure consistency and clear boundaries from the other themes. Each theme was then defined and named to accurately capture its meaning in relation to the research questions. Finally, the themes and representative quotes were compiled into a structured narrative for presentation in the results section. Since all interviews were conducted in Swedish, the coding process was carried out in the original language to preserve meaning. Quotes selected for inclusion in the results section were translated to English.

3.5.2 Quantitative Analysis

Quantitative analysis was applied to the data collected through ERP extraction, scrap bin measurements and coil offcut measurements. This method was selected since it enables scrap generation to be quantified and compared across process stages using UTL as a normalised performance indicator, making it well suited for establishing a process-level baseline and identifying patterns in the production data independent of production volume fluctuations. Quantitative methods are particularly appropriate when the research objective requires measurable and reproducible results that can be systematically compared (Bryman, 2016).

The analysis was conducted in several steps. The raw ERP data was first cleaned and structured in Excel to remove incomplete entries and standardise formatting. UTL percentage was used as the primary performance indicator, provided directly in the ERP data. Where scrap weight was needed for comparison with manually collected data, it was derived by combining UTL with the sheet area and material surface density. Scrap measurement data was compiled separately and aggregated by process stage and time to enable comparison with the ERP derived data.

Coil cutting scrap was estimated by calculating the average coil offcut length per coil change from the recorded measurements. This average was then combined with the corresponding coil width and material surface density to estimate the scrap generated at each coil change. This weight was then allocated across all articles produced within the same batch, defined as articles produced between coil changes each production day. By adding the distributed scrap weight to each article's sheet weight, a UTL percentage accounting for coil cutting scrap could be calculated on a per-article basis. This calculation assumed that the operators sequence all articles of the same coil width consecutively, minimising the number of coil changes performed each day. This behaviour was consistently observed during the observation sessions but represents an assumption rather than a verified rule. As a result, the coil cutting scrap estimate carries a degree of uncertainty and should be interpreted accordingly. To visualise and interpret the compiled data, run charts, and histograms were applied. These tools were selected as they are commonly used within the Measure phase of DMAIC to establish process baselines, visualise variation and identify potential sources of scrap generation (Basu, 2011).

3.6 Validity and Reliability

The quality of this study has been evaluated against four criteria proposed by Yin (2018) for assessing the rigour of case study research: construct validity, internal validity, external validity and reliability. As seen in Table 3.2, each criterion was addressed through specific measures. Construct validity concerns whether the study measures what it claims to measure (Yin, 2018). This was addressed through the use of multiple sources of evidence, as recommended by Yin (2018). Scrap generation was measured through four independent sources: ERP data, manual scrap bin measurement and qualitative data from interviews and observations. This enables triangulation and allows for cross-validation of the findings. Where results from different sources aligned, confidence in the measurement was strengthened. Where results misaligned, ERP data was treated as the primary source of truth due to its automated and traceable nature. In addition, the structured observation protocol ensured that observations were conducted systematically rather than unstructured, and the semi-structured interview format ensured that key topics were covered across all respondents.

Internal validity is primarily relevant to causal or explanatory studies seeking to establish relationships between variables (Yin, 2018). As this study is descriptive and exploratory in nature, internal validity is not a primary concern. The study does not seek to determine causal relationships between variables but to measure and describe the distribution of scrap generation across process stages. However, the DMAIC structure forced a sequential logic during the course of the study, reducing the risk of drawing conclusions that are not grounded in collected data.

External validity concerns the degree to which the findings of this study can be generalised beyond the immediate case (Yin, 2018). Following Yin (2018), this study relies on analytic generalisation rather than statistical generalisation. The basis for this claim is not that the case is representative of a broader population, but that the contextual conditions present in this case, namely MTO production, hybrid batch-line process structure, and limited automation, are not unique to this facility. These conditions are typical of a broad category of small-to-medium sheet metal manufacturers operating in engineer-to-order or make-to-order environments, where process-level measurement infrastructure is rarely fully developed (Frohm et al., 2008; Shahbazi et al., 2016). The measurement approach, the application of DMAIC as a framework for scrap identification, and the identified relationship between automation level and measurement system design are therefore analytically generalisable to manufacturers operating under similar conditions. The specific scrap volumes and distributions identified in this study are context-dependent and cannot be transferred directly, but the mechanisms through which scrap becomes invisible to existing reporting systems, and the measurement design required to address them, are likely to appear in similar ways in comparable production contexts.

Reliability concerns whether this study can be repeated by another investigator and yield the same results (Yin, 2018). This was addressed in several ways. Data collection procedures were documented in detail, including observation protocol, interview guides and ERP extraction parameters. All interviews were audio recorded and transcribed to ensure that the analysis was based on accurate representations of respondent statements rather than on researchers' memory of them. The thematic analysis followed the 6-step process outlined by Braun and Clarke (2006), providing a transparent and reproducible analytical procedure. ERP data was stored in its raw form to enable verification between data processing, and manual bin measurements were logged separately for each bin location and with timestamps to reduce the risk for measurement error. Together, these measures reduce the risk of researcher bias affecting the data and allows the study to be audited and reproduced.

Table 3.2
Research quality criteria and corresponding measures

Criterion	Measure taken
Construct validity	Multiple data sources (ERP, manual weighing, observations, interviews) enabling triangulation
Construct validity	Structured observation protocol and semi-structured interview format
Internal validity	DMAIC sequential structure preventing conclusions not grounded in collected data
External validity	Analytic generalisation to similar MTO sheet metal contexts with hybrid batch-line structures
Reliability	Detailed documentation of data collection procedures
Reliability	Audio recording and transcription of all interviews
Reliability	Thematic analysis following Braun and Clarke (2006) six-step process
Reliability	ERP data stored in raw form for verification
Reliability	Manual bin measurements logged with timestamps and bin location

3.7 Use of Artificial Intelligence

Generative AI, specifically Claude (Anthropic) and Microsoft Copilot, was used at several stages of this study. Its use is disclosed here in accordance with the guidelines set by the School of Engineering at Jönköping University. For tasks involving sensitive production data, including ERP extracts and scrap measurement records, the case company's data handling guidelines were followed. Where AI tools were used in this context, this was limited to Microsoft Copilot accessed through the case company's enterprise licence. Claude was exclusively used for non-sensitive tasks described below, and no production data or company-specific figures were submitted to Claude at any stage of the study.

The primary use was in writing and language refinement. Where AI was used for writing, the authors drafted all text independently before submitting it to AI for linguistic improvement. This included restructuring of sentences, improving clarity and correcting grammar. AI was also used to support the reading process when engaging with theoretical sources. This was limited to summarising or refreshing the content of sources that had already been read and located by the authors, to aid in recall and overview. Scopus AI was additionally used to identify relevant literature during the search process. All sources were independently verified by the authors regardless of how they were identified. For data processing, AI was used to translate analytical procedures and logical operations, developed by the authors, into functional formulas and Power Query transformations in Excel. The underlying logic and calculations were defined by the authors prior to implementation. AI was also used to support the transcription and qualitative coding of the semi-structured interviews. The resulting transcripts and codes were reviewed and validated by the authors against the original recordings to ensure accuracy. In all instances, AI was used as a tool rather than a source of analytical conclusions or empirical data. The authors take full responsibility for the content, interpretations and conclusions presented in this study.

3.8 Ethical Considerations

The ethical considerations were incorporated into the research process to ensure the integrity of the study and the protection of participants. Prior to each interview, the participants were informed about the purpose of the study, the expected duration, how the data would be used and their right to decline the participation. Verbal consent was obtained from all participants before the interview began. All participants volunteered to participate. All interviews were audio recorded with the consent of each participant. The recordings were used solely for transcription purposes and were not shared with any third party. To protect participant anonymity, no personal names were used in the study. Respondents are referred to exclusively by their professional role, such as team leader, operator or quality engineer. This ensures that individual statements cannot be traced to specific individuals.

The study was conducted in collaboration with the case company, which maintained an active role in reviewing the report throughout the writing process. Each iteration of the report was reviewed by a company representative to ensure that confidential production data was not leaked to the public. Specific scrap weights and other sensitive operational figures were excluded from the final report to comply with the company's confidentiality requirements.

No formal ethical review was required for this study, as it did not involve sensitive personal data, vulnerable populations or any form of intervention. The study focused on production processes and material flows rather than on individuals, and all data was collected in a workplace context with informed and consenting participants. It is acknowledged that the findings and improvement suggestions presented in this study could, if implemented, have an impact on the organisation at the case company, including changes of roles or staffing levels. However, this study is limited to identifying and measuring sources of scrap and proposing potential improvements. Decisions regarding implementation fall outside the scope of this study and remain with the case company.

4 Results

This chapter presents the empirical findings of the study, structured around the production system, measurement baseline and scrap variation findings. The chapter begins by presenting the raw empirical findings from the data collection methods employed during the study, followed by the application of DMAIC tools to structure and visualise the collected data.

4.1 Production System Overview

The Define phase established the scope and boundaries of the study by mapping the production flow, identifying where scrap is generated and capturing process-specific knowledge. The findings were structured using a SIPOC diagram and process flow diagram, which together defined the process boundaries and scrap output streams that formed the basis for data collection in the subsequent Measure phase.

4.1.1 SIPOC Diagram

A SIPOC diagram was developed to map the panel production line at a high level. In line with Basu (2023), the SIPOC identifies the main suppliers, inputs, process steps, outputs and customers of the system, and establishes the process boundary before moving into detailed measurement. The resulting diagram is shown in Figure 4.1.

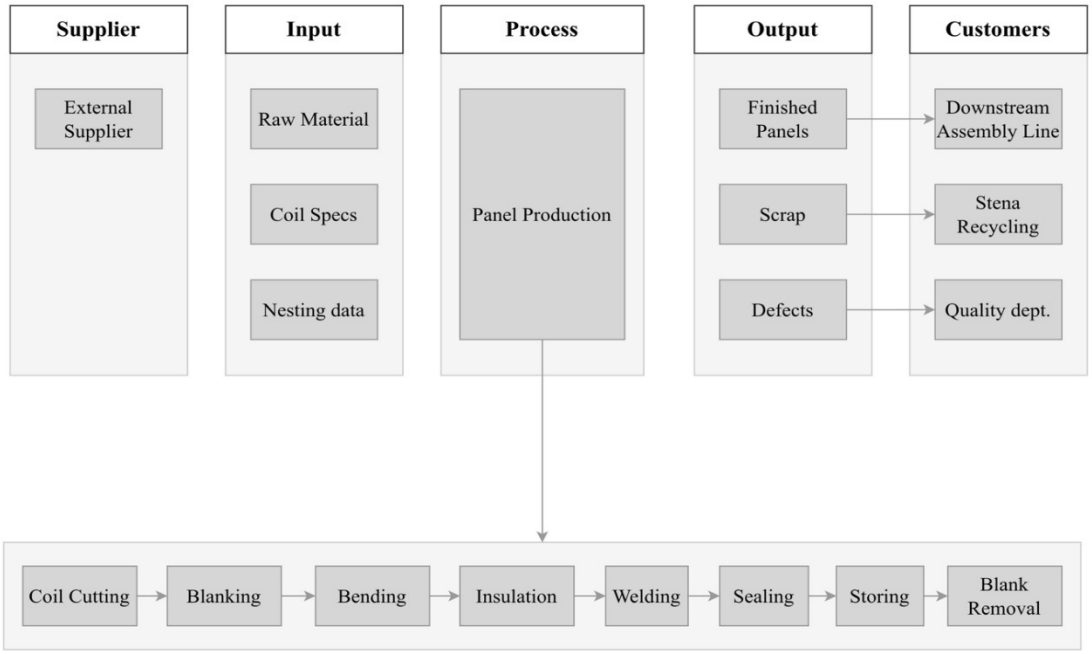


Figure 4.1
SIPOC diagram of the insulated panel production line

The process receives its material from external suppliers that delivers coiled sheet metal to the factory. Once on site, the coils are held in a coil inventory buffer before entering the line, making raw coiled steel and in-plant inventory the two inputs to the process. From there, the production sequence consists of eight steps. The coil is first cut into sheets of batch specified dimensions, after which the sheets pass through the punching machine where the required hole patterns and panel profile geometry are produced. The punched sheets are then bent into the panel profile shape and loaded with rock wool insulation to form the panel core assembly. The panel frame is then welded to enclose the assembly, sealing strips are applied to the panel edges, and the completed panels are transferred to the automated inventory system. Where remaining blanks have not been removed in earlier stages, they are punched out manually in a post-processing step before the panel leaves the line.

The line produces three outputs. The primary output is the finished panel sub-assembly passed to the internal production line for integration into ventilation units. The secondary outputs are metal scrap in the form of offcuts and trim loss. Accordingly, the SIPOC identifies two customers: the internal production line as the primary customer receiving finished panels, and Stena Recycling as the secondary customer receiving the metal scrap.

4.1.2 Process Flow Diagram

A process flow diagram was developed for the insulated panel production line and is presented in Figure 4.2. The diagram maps the complete material flow from goods receiving to panel output, including all decision points, scrap output streams and the transition between the batch and line production segments. Solid arrows represent the main material flow, and dashed lines represent scrap outputs and conditional branches.

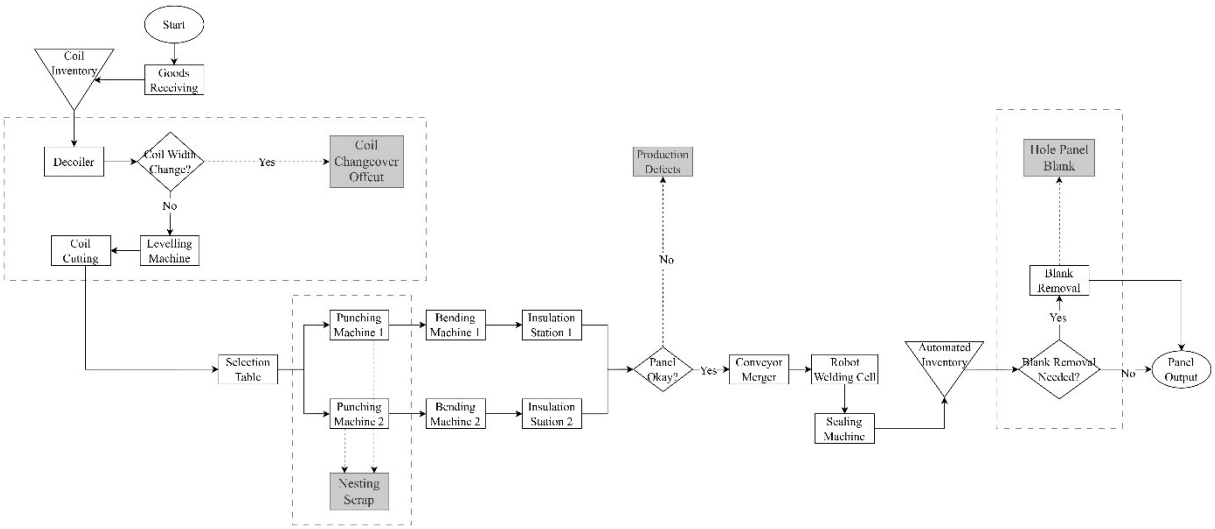


Figure 4.2
Process flow diagram of the insulated panel production line

The flow begins at the start terminal, which leads to goods receiving, where incoming coils are registered and placed in a coil inventory buffer. From coil inventory, material is fed to the decoiler, where a coil is always loaded and ready for production. Before processing begins for the next order, the operator checks whether the coil width currently loaded in the decoiler matches the width required by the upcoming

order. If the width matches, production continues with the coil already in the machine without interruption. If the required width differs from the loaded coil, the coil is changed before processing continues. Following the coil change, the new coil is fed through the levelling machine, which flattens the strip and straightens any curvature from coiling. It is at this stage, after levelling the new coil, that Coil changeover offcut is generated, as the leading portion of the newly loaded coil is cut away before dimensional quality is sufficient for production. The levelled strip then continues to the coil cutting machine, where sheets are cut to the specified dimensions required by the current order. The cut sheets accumulate on the selection table, which feeds the downstream production line. The stages from coil inventory through the selection table constitute the batch processing segment of the production system, in which material is prepared and accumulated before being released to the line.

From the selection table onwards, the system transitions into a line production segment, where panels move sequentially through fixed process steps. The line segment consists of two configured production paths: Path 1, comprising “Punching Machine 1”, “Bending Machine 1” and “Insulation Station 1”, and Path 2, comprising “Punching Machine 2”, “Bending Machine 2” and Insulation “Station 2”. Only one of these paths is active at any given time; the active path is selected based on the order being processed. Both punching machines generate nesting and geometric offcuts during the punching operation, which exit the process as “Design Scrap 1”, indicated by dashed lines from both machines to a shared scrap output.

After the insulation station on whichever path is active, the flow reaches the second decision point, which assesses whether each panel meets quality requirements. Panels that do not conform are removed from the flow and recorded as “Production Defects”. Panels that pass the quality check continue to the “Conveyor Merger”, where the output of both production paths is consolidated into a single flow regardless of which path was active.

From the “Conveyor Merger”, panels proceed sequentially through the “Robot Welding Cell”, where the panel assembly is welded, and then through the “Sealing Machine”, where sealing strips are applied. After sealing, panels are transferred to “Automated Inventory”, which acts as a storage buffer before the final processing stage.

From “Automated Inventory”, the flow reaches the third decision point, which determines whether blank removal is required. Blank removal applies to panels where the punched blank geometry exceeds the capacity of the punching machine to eject automatically, leaving the blank still attached to the panel. If no blank removal is needed, panels proceed directly to Panel Output. If blank removal is required, panels are routed to the “Blank Removal Station”, where the remaining attached blanks are manually punched out. The material removed at this station exits as “Design Scrap 2”. After blank removal, panels continue to “Panel Output”, which is the final terminal of the process flow.

4.2 Process Observations

The observation findings from the three sessions described in section 3.4.1 are summarised in Table 4.1.

Table 4.1
Summary of structured observations across three production segments

Category	Observation A: Coil handling	Observation B: Punching, bending, insulation	Observation C: Blank removal
Process steps observed	Coil unroller, coil leveller, rotating table, coil magazine, coil cutter	Selection table, punching, bending, insulation station	Automated storage system, blank removal
Scrap-generating steps	Coil cutter	Punching	Blank removal
Automation level	Semi-automatic; operator actively adjusts process and inspects cut sheets	Punching and bending fully automated; insulation station: manual	Semi-automatic; Binar feeds panels to operators
Type of scrap	Process scrap (coil changeover offcut)	Design scrap (nesting and geometric residuals)	Design scrap (blanks removed from hole panels)
Scrap pattern	Intermittent; produced during changeover phase	Continuous	Intermittent
Machine settings	Variable; leveller parameters adjusted at each coil change and during operation	Static	Static
Defect identification	Visual inspection of cut sheets by operator	Visual inspection of panels at insulation station	Visual inspection
Existing scrap measurement	None	Logged automatically in ERP system (UTL %)	None

The observations revealed several findings that informed the definition of process boundaries and the scope of the measurement approach. Scrap was observed at three distinct points within the production system. At the coil cutting machine, process scrap was generated during each coil changeover, when the first meters of new coil are trimmed and discarded due to surface damage. At the punching machines, design scrap was generated continuously because of the nesting, where trimmings were separated from the sheet to achieve the geometry needed for bending. At the blank removal station, design scrap was

produced in the form of blanks punched out from the hole panels. Hole panels are an integral part of an AHU, allowing for air passage through the unit, generally as a part of an intake or a mounting point for a fan. Figure 4.3 shows a nesting software view of a hole panel variant. The black areas represent trim losses generated during punching, which the nesting system logs automatically in the ERP. The dark grey area represents the internal blank, which is manually removed at the blank removal station but generates no corresponding ERP record, meaning it is physically scrapped but invisible in the system data.



Figure 4.3
Nesting software view of a hole panel

Comment. ERP-logged trim loss (black) and the unrecorded internal blank (dark grey)

The observations also revealed a clear difference in process variability between the upstream and downstream segments. In the coil handling segment, machine settings were not static. The operator adjusted leveller parameters at each coil change and could make further adjustments during operation to accommodate for variations in coil properties. These operator-dependent adjustments introduce a source of variation that is not present in the downstream punching and bending stages where the level of automation is higher. Of the three stages, only the punching stage had an existing measurement system in place, where UTL and other parameters were logged automatically in the ERP system on a per-article basis. The coil handling and blank removal segments had no active scrap measurement. The observations were conducted within the process boundaries established through the SIPOC diagram and process flow, with each session targeting the production segments identified there.

4.3 Interview Findings

Semi-structured interviews were conducted to capture process-specific knowledge and challenges. Compilation of the interview data resulted in four themes: process-level measurement challenges, sources of scrap generation, sources of variation and improvement opportunities. The themes and their key sub-themes are visualised in Figure 4.4. The themes reflect the theoretical distinctions in section 3.4.1, with theme 2 drawing on the categorisation of material scrap types and their origins, theme 3 on the distinction between chance causes and assignable causes of variation, and theme 4 on continuous improvement principles. Themes 1 and 2 primarily address RQ1, theme 3 addresses RQ2, and theme 4 addresses RQ 3.

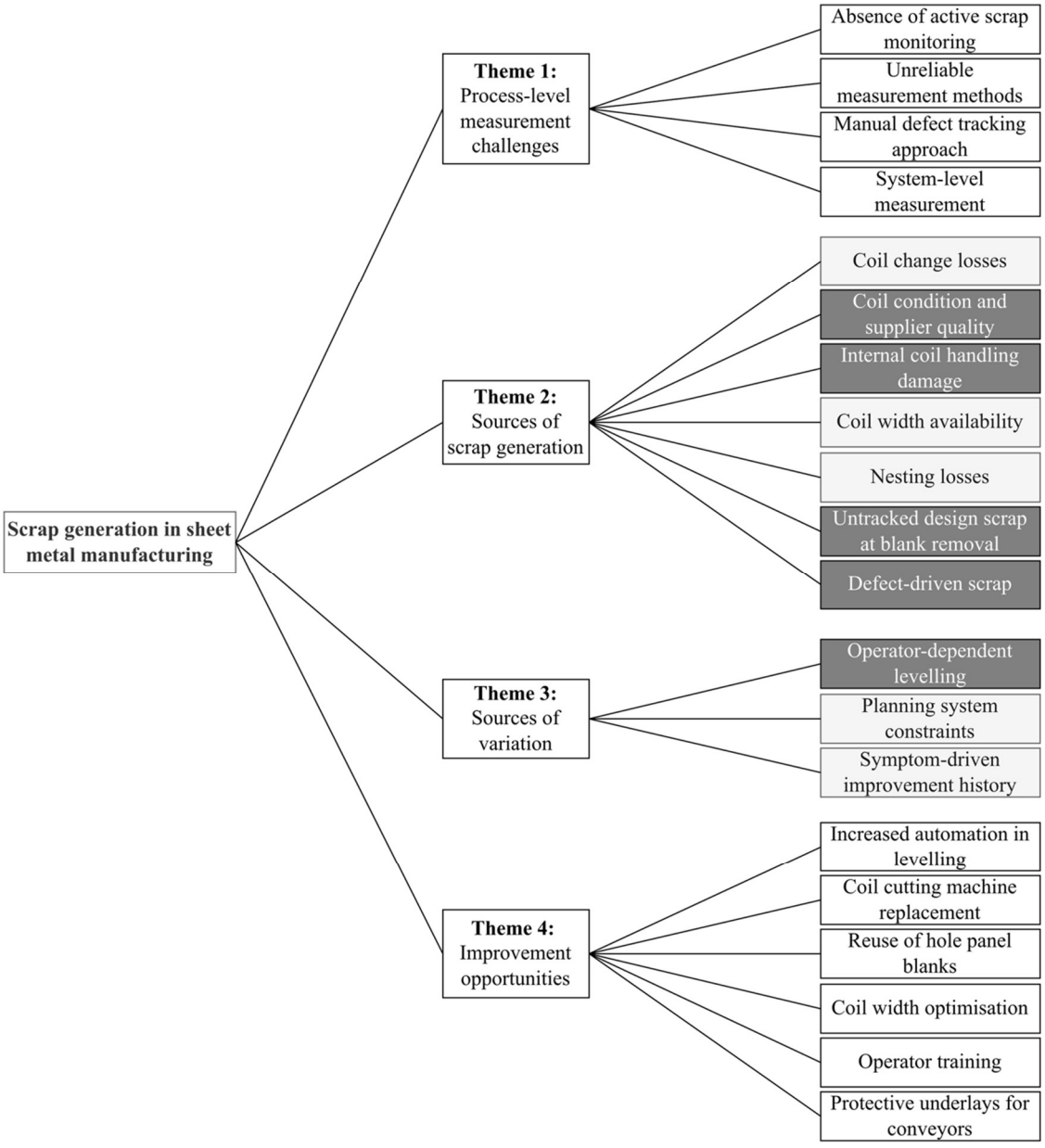


Figure 4.4

Thematic tree diagram of scrap generation in sheet metal manufacturing.

Comment. In themes 2 and 3 dark-filled nodes indicate assignable causes; white-filled nodes indicate chance causes. Theme 3 indicates challenges and theme 4 indicates improvement opportunities.

4.3.1 Theme 1: Process-Level Measurement Challenges

A recurring finding across all respondents was the absence of a deployed strategy for performance measurement system for process-level scrap. Scrap was described as something that had historically been disposed of without being quantified or tracked at the individual process stage. As the interviewed team leader noted: *"scrap monitoring is not actively followed up… there is no active monitoring, scrap is just thrown away without being measured."* This was reinforced by the team leader's description of how scrap has historically been handled: *"we have never had any concerns; we just throw it in the container… and then we don't know how much has been discarded"*

Where measurement did exist, it was described as unreliable or imprecise. The team leader noted that the current weighing method is based on estimation rather than direct measurement *"We have weighed the bins when they are completely full… if they are filled less then we take away some… so it's not 100% accurate there."* A further complication was identified in that the recorded bin weight includes the empty bin itself: *"What is included is calculated together with the empty bin itself, so you have to subtract from each row."*

Recent initiatives had begun to address the lack of measurement. Quality engineer 2 described how a Six Sigma project had recently introduced more active monitoring in another production line: *"until now we have started doing this… previously we have not followed up on this so much in terms of UTL."* However, measurement remained on an aggregate basis rather than process-level: *"we look more at the overall picture"*. The same respondent also highlighted a fundamental challenge with percentage-based performance measurements: *"even if it sounds like 90% UTL is very good, given the volume you run, it becomes a lot in the end."*

4.3.2 Theme 2: Sources of Scrap Generation

The coil cutting stage was consistently described as the primary source of material scrap, with observed losses occurring at both the start and end of each coil of metal. The team leader described how losses occurred at the start of every coil: *"on each coil we cut almost five to six meters and discard it."* This was confirmed by the operator, who noted that the losses occur at both ends of the coil: *"it is always the first metres at the beginning of the coil that are a bit scratched… you always have to discard some meters at the beginning."* The amount discarded was described as variable: *"sometimes it looks really good after three- four meters, sometimes you have to discard maybe seven-eight-ten."*

Beyond the losses at coil changes, the condition of the coils themselves was identified as a source of scrap. The team leader noted that the supplier quality affected the scrap generation: *"we have two different suppliers with the coils that are delivered to us, one of them produces more scrap."* Internal material handling was also highlighted as a cause of material damage. The operator described how improper storage had led to rust damage: *"rust appeared due to internal handling… it was stored outside and was not protected well enough."* Repeated coil handling due to coil changes was further described as generating additional scrap each coil mount: *"you back it and then the day after you lift it again and then you discard again just because there had become a few scratches."*

At the punching stage, trim loss was consistently described as a significant scrap source. Quality engineer 2 identified these as the largest single contributor*: "I believe it is the nesting losses that generate most scrap… process scrap as we loosely call it."* The team leader illustrated how coil width mismatches can drive yield losses*: "you have produced a thousand panels, but the panel should only be 800mm, 200mm becomes scrap … it should have been produced on a narrower coil."* The engineer

further clarified that the punching machine also trims the remaining edge of the sheet: *“it is not only the small holes we cut in the sheet but also the offcut, so to speak. And that is included in nesting efficiency.”*

The available range of coil widths was described as a built-in limitation. Quality engineer 2 noted that the current rotating coil feeding system to the coil leveller limits the number of available widths: *"today we have a rotation system... we can rotate between four different coils. But if you could take for example 12, then you could have many more widths."* The available widths are further constrained by standardised market offerings: *"usually when you buy a coil you buy standard widths because it is much cheaper."* The nesting logic itself was also described as a source of variation, as it is an internally built system rather than an optimised program: *"it is not a program that nests for us... it would be good if you could somehow optimise the different types of combinations."*

A fourth category of scrap, largely invisible in the existing data, was identified at the blank removal stage. Quality engineer 2 described how hole panels generate design-inherent scrap: *“there, scrap is generated in the form of punching out hole panels… but this is not included in the nesting scrap, so you don’t see it in the data.”* Given that every AHU produced contains at least two-hole panels for air to be able to pass through, this represents a systematic and untracked source of material loss.

4.3.3 Theme 3: Sources of Variation

Variation in scrap generation was attributed to chance causes, representing systemic causes and sources of variation inherent to the process, and assignable causes, representing identifiable sources of variation that can be traced to specific events and factors. The planning system represented a chance cause of variation. As the team leader explained: *"we cannot run any orders today that are due on Friday... it is controlled by automation."* This prevents operators from grouping orders by coil width to minimise coil changes, a practice that was identified as an effective way to reduce scrap: *"you could have reduced the scrap by running more with the same coil after each other."* Manual width adjustment was theoretically possible but was described as avoided in practice: *"it is very time-consuming work for us... people do not dare to go in and change."*

On the human side, operator skill was identified as an assignable cause of variation in scrap generation at the coil levelling stage. Quality engineer 2 noted: *"how you level is very person-dependent. Some level well and some level poorly"*, estimating that differences in levelling quality could account for *"maybe two metres per sheet."* The operator confirmed that machine knowledge is largely built through on-the-job experience: *"the training is not on that specific machine because you cannot learn it anywhere else... it is only there."*

Previous improvement efforts were described as having addressed symptoms rather than root causes. Quality engineer 1 noted: *"historically, you have always made point interventions where you think most quality defects occur."* Despite repeated attempts, defect rates had proven difficult to sustain at lower levels: *"sometimes you manage to bring it down a few percentage points, but shortly after it jumps back up again as the transport conveyors wear over time."*

4.3.4 Theme 4: Improvement Opportunities

Respondents proposed improvements targeting the coil cutting stage. At the coil cutting stage, machine replacement was identified as the most impactful measure, representing a kaikaku-type improvement through fundamental redesign of the process. The team leader noted that management had visited a

comparable facility abroad: *"our managers have been down to Germany and seen how their process works... they do not have one like ours."* Quality engineer 2 confirmed that machine replacement was being considered: *"there are plans to replace that machine over there in P10, but there are plans to replace this one too eventually."* While acknowledging the cost, the same respondent noted that it would likely be financially justified: *"it is expensive, but if you calculate it, I do not think it is that much in the long run."*

Coil width optimisation was identified as a complementary measure with significant potential. Quality engineer 2 described it as the most effective option available: *"the coil width is the easiest, or most effective thing to look at."* Increasing the number of available coil widths through an expanded rotation system was proposed to reduce trim loss without requiring significant capital investment in new equipment, representing kaizen-type improvements achievable within the existing process structure. Operator training was raised as a lower-cost short-term measure for reducing variation at the levelling stage: *"training, really"*, with automation of the levelling process identified as the longer-term solution*: "if you had something that could level by itself... then you would naturally get much less scrap."*, which would be a kaikaku intervention in the production process.

4.4 Scrap Measurement and Baseline

During the Measure phase, quantitative data was collected to establish a process-level scrap baseline across the production stages. The phase produced scrap distribution data across the three process stages, coil offcut measurements and UTL data, together forming the empirical basis for the subsequent analysis. The following sections present the statistical tools applied to identify variation patterns in scrap generation across the measurement period.

4.4.1 Scrap Distribution by Data Source

Table 4.2 presents the scrap distribution across production stages as recorded by the ERP system and manual bin weighing. The punching machine represented the dominant scrap contributor across both sources. The blank removal station was not captured in the ERP system and is therefore absent from that distribution. Of the recorded bin emptyings at the punching station, 75% occurred when the bin was at approximately full capacity, with the remaining 25% recorded at below full capacity.

Table 4.2
Scrap distribution across production stages by data source

Production stage	ERP (%)	Manual weighing (%)	Manual weighing excl. blank removal (%)
Punching machine	74	61	75
Coil cutting machine	26	20	25
Blank removal station	-	19	-

Note: Manual weighing excl. blank removal redistributes the manual weighing figures across the two remaining stations, excluding the blank removal station to enable direct comparison with ERP data.

4.4.2 Coil Offcut Measurements

Figure 4.5 presents the coil offcut length at each coil change, normalised against the overall average. A total of 14 measurements were recorded across four production days, capturing the length of the material trimmed as offcuts and discarded at the start of each new coil. Offcut scrap was generated at every coil change without exception. In addition to the start-of-coil measurements, the offcut length at the end of coil was also recorded when coils were fully consumed. However, only three such instances occurred during the measurement period, which was considered insufficient for further analysis.

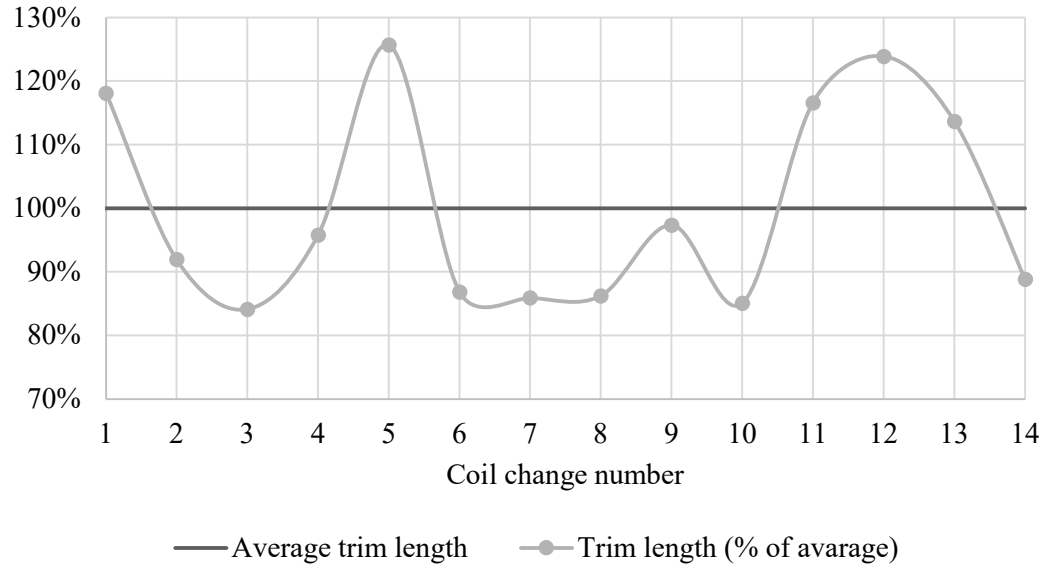


Figure 4.5
Coil change trim length relative to average across 14 recorded coil changes

4.4.3 Run Chart

Figure 4.6 presents the weekly average UTL for the punching stage and the coil cutting stage over the measurement period. Both series remain within a relatively narrow band throughout the period, with no sustained upward or downward trends visible.

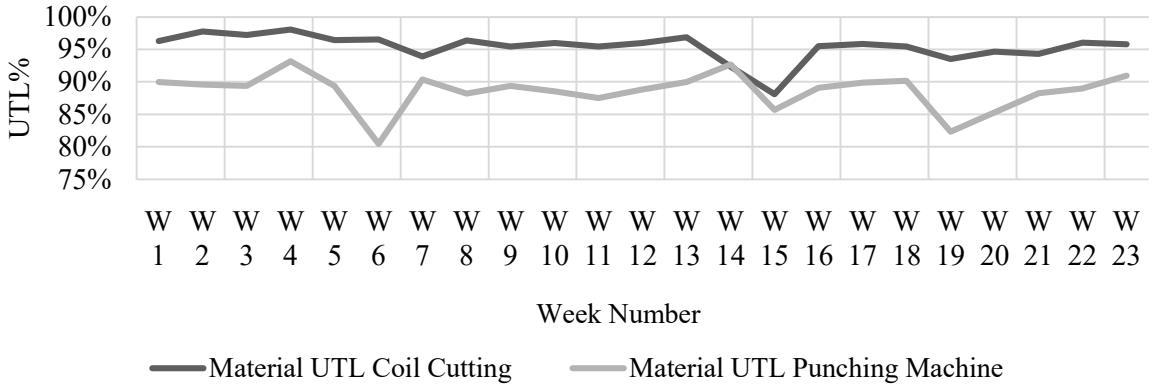


Figure 4.6
Run chart for average material utilisation (UTL)

Comment. Only intact weeks shown

4.4.4 Histogram

Three histograms were constructed from the ERP dataset to illustrate the distribution of UTL across the production system. Each histogram grouped historically produced articles into 5-percentage-point bins based on their UTL. Figure 4.7 presents the distribution of UTL for the punching stage, reflecting the efficiency of the nesting process. The distribution is right skewed, with the highest concentration of articles in the 85%-100% range. A smaller number of articles is distributed across the lower utilisation bins with relatively few articles falling below 70%.

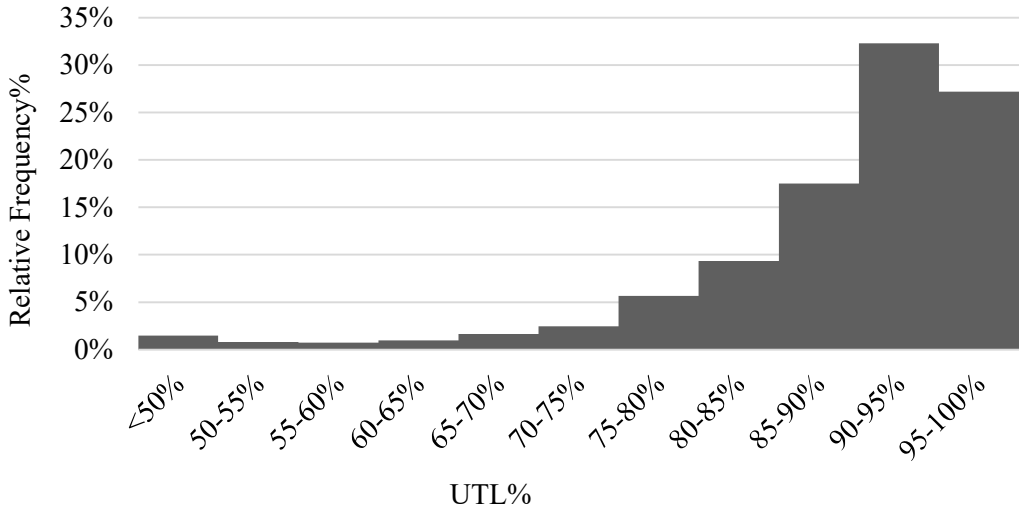


Figure 4.7
Distribution of material utilisation at the punching stage

Figure 4.8 presents the distribution of UTL when accounting for offcut scrap allocated across articles produced within each batch. The distribution follows a similar right-skewed pattern, with an even stronger concentration in the 90%-100% range. The lower bins contain relatively few articles

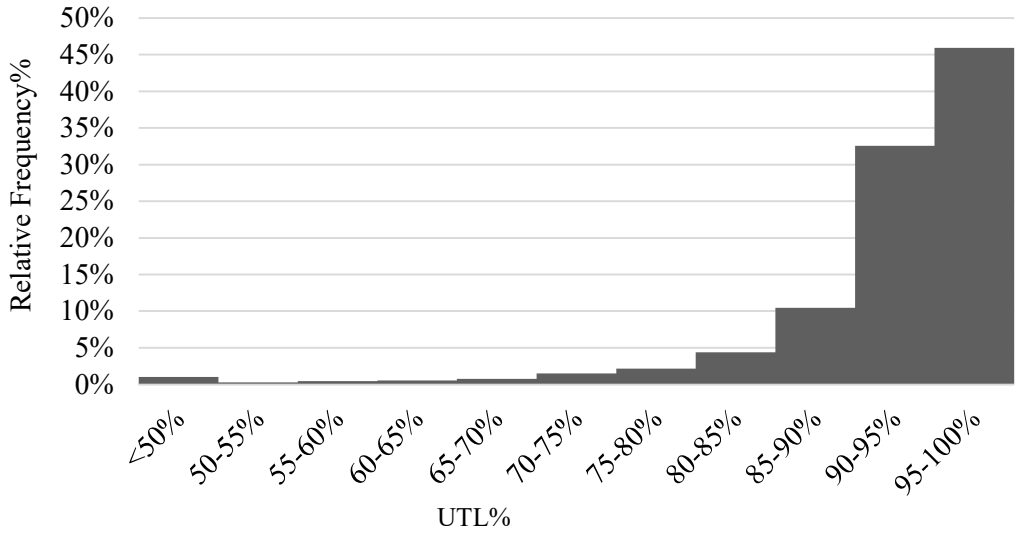


Figure 4.8
Distribution of material utilisation at the coil cutting stage

Figure 4.9 presents the combined UTL across both the punching and coil cutting stages. The distribution is right skewed, with the highest concentration of articles in the 85%-95% range.

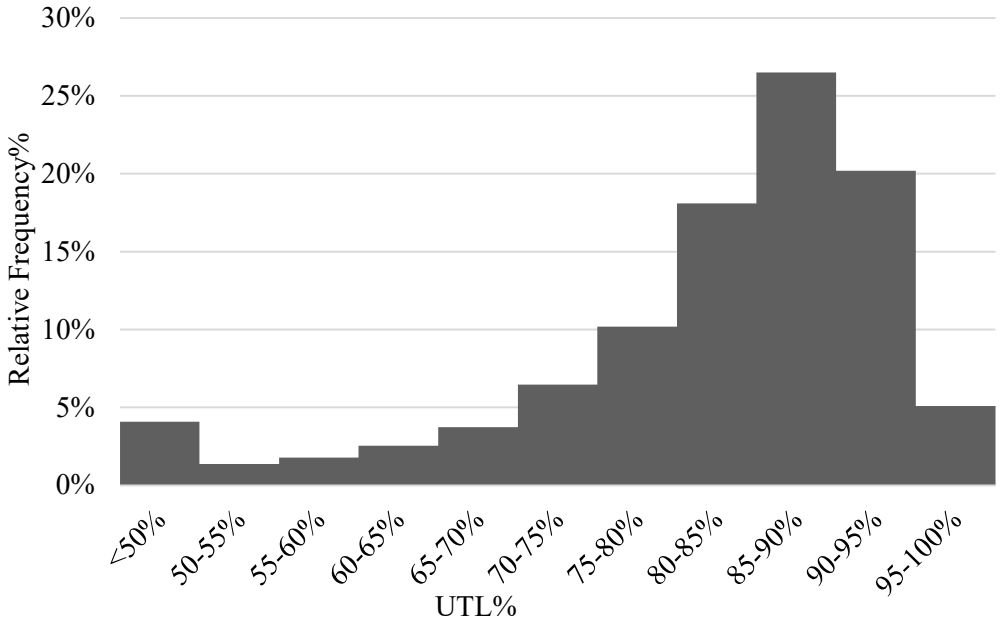


Figure 4.9
Distribution of material across the total production system

4.4.5 Cause and Effect Diagram

A cause-and-effect diagram was constructed to identify and organise the perceived causes of low UTL in the panel production line. The diagram applies the 6M framework, with six main cause categories: Machine, Method, Material, Manpower, Measurement, and Environment, with "Low UTL" as the stated effect. An Environment branch is included in the diagram structure but contains no identified causes. The resulting diagram is shown in Figure 4.10.

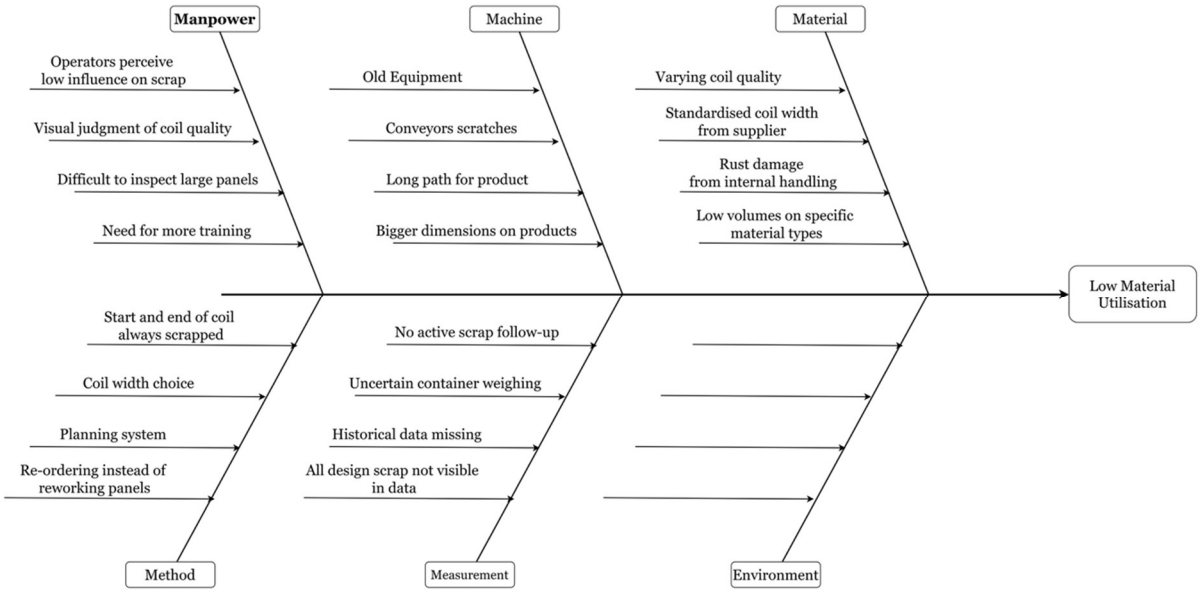


Figure 4.10
Cause and effect diagram identifying perceived causes of low UTL

In the Manpower category, four causes were identified. Operators perceive low personal influence over the amount of scrap generated. Coil quality at start is assessed by visual judgement. Large panels are difficult for operators to inspect fully. A need for more training was also identified. The Machine category contains three causes: old equipment, a long mechanical path for the product through the line and increasing product dimensions over time that the equipment has not been adapted to accommodate. Under Material, four causes were recorded: varying coil quality between suppliers, standardised coil widths from the supplier that limit selection, rust damage on coils from internal handling and low production volumes on specific material types leading to lower utilisation. In the Measurement category, four causes were identified: no active scrap follow-up, uncertain container weighing, missing historical data and design scrap not being fully visible in the available data.

5 Analysis

This chapter analyses the empirical findings presented in chapter four against the theoretical framework. The chapter is structured around the three research questions, each corresponding to one or more DMAIC phases. RQ1 draws on the Define and Measure phases to establish how scrap and UTL can be defined and measured at process-level. RQ2 addresses the Analyse phase by identifying and categorising the root causes of variation in UTL and the process-level measurement challenges, drawing on the cause-and-effect diagram and validated against the quantitative data. RQ3 addresses the Improve phase by proposing and assessing the feasibility of scrap reduction measures grounded in the identified root causes. The connection between research questions and DMAIC phase is visualised in Figure 5.1 below.

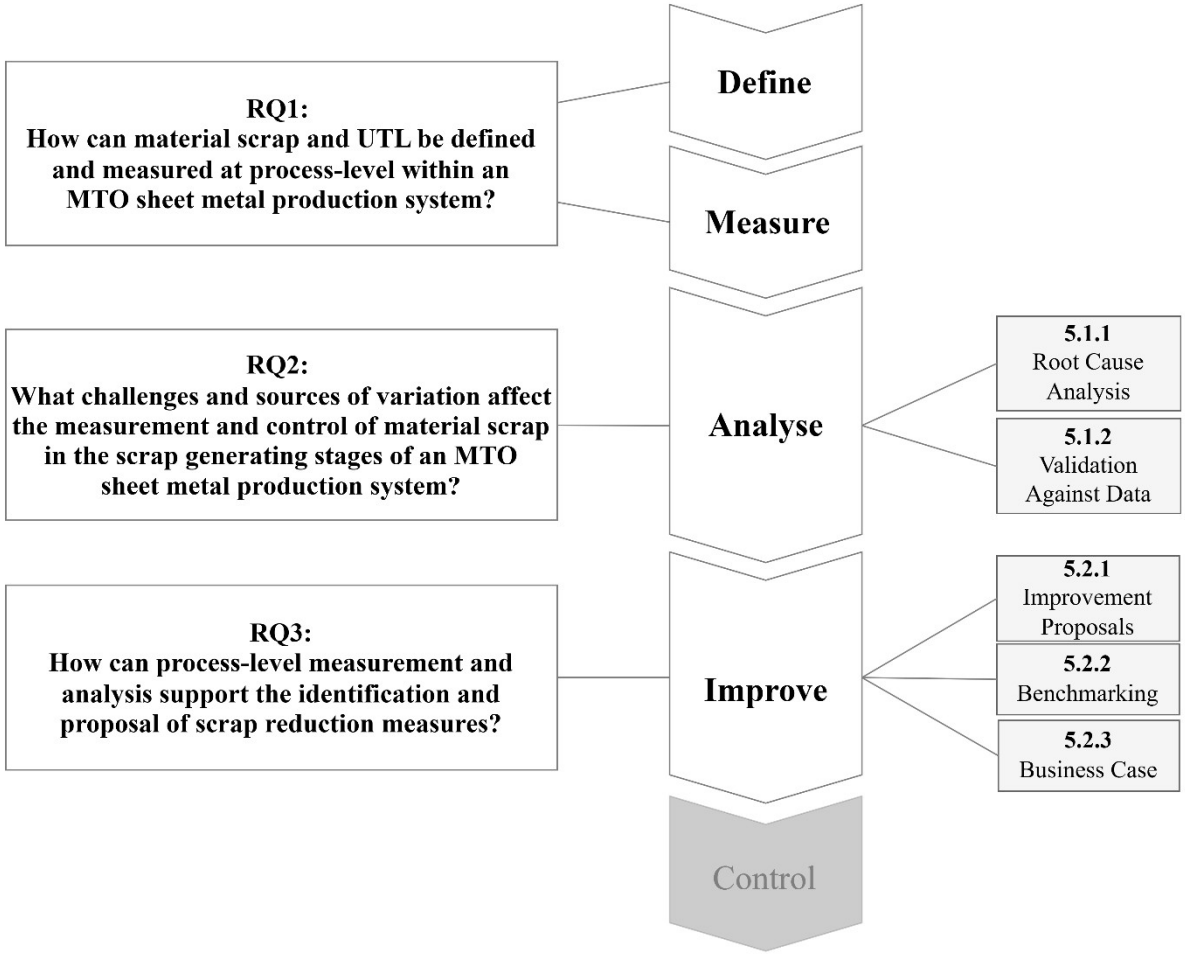


Figure 5.1
Connection between RQs, DMAIC phases and corresponding analysis

5.1 RQ1: How can material scrap and UTL be defined and measured at process-level within an MTO sheet metal production system?

RQ1 is addressed through the Define and Measure phases presented in chapter 4. The theoretical framework predicts that process-level scrap measurement in MTO environments is complicated by three overlapping conditions: order-driven variability in material specifications, mixed automation levels across production stages, and the tendency of existing reporting systems to aggregate losses at line or plant level rather than by stage (Frohm et al., 2008; Neely et al., 1995; Shahbazi et al., 2016). All three conditions were encountered empirically. No process-level measurement existed prior to the study, scrap was disposed of without measurement, and the ERP system recorded losses at the punching stage only, leaving two additional scrap-generating stages unaccounted for. This reflects what Neely et al. (1995) identify as the core limitation of aggregate measurement: knowing how much scrap a system produces tells a manager nothing about where it comes from or what to do about it.

Before a baseline could be established, the scrap-generating stages and their losses had to be defined. The Define phase identified three distinct scrap flows: coil changeover offcut, trim loss, and blank removal scrap. Under Datar and Rajan's (2017) typology, all three qualified as normal spoilage, since each arose from the process design or product geometry, not from process deviations. This matters because it rules out operational correction as a viable reduction strategy for any of the three flows, but also had implications for measurement design. Because each flow was generated through a different physical mechanism at a different point in the production system, no single data source could cover all three. At the punching stage, UTL was logged automatically per article by the ERP system, making it a latent KPI that was available but never extracted or acted on. At the coil cutting stage, scrap weight had to be estimated by combining manually measured offcut lengths and ERP-derived coil change amounts. At the blank removal station, manual bin weighing was the only available data source, with no ERP record at that stage. Montgomery (2008) establishes that instrument, method, operator, and measurement conditions must collectively produce reliable results before data can serve as an analytical baseline, and the variation in data availability across stages meant this had to be evaluated differently for each. Since a formal Gauge R&R was not feasible without a defined instrument or stable conditions, cross-validation between independent sources served as the primary consistency check, as examined further in section 5.2.

5.2 RQ2: What challenges and sources of variation affect the measurement and control of material scrap in the scrap generating stages of an MTO sheet metal production system?

This section addresses RQ2 by examining the root causes of variation in UTL and the process-level measurement challenges identified through the empirical findings in chapter 4. The chance and assignable cause framework is used to categorise the identified sources of variation, as this distinction determines both the nature of each challenge and the type of intervention required to address it. The categorisation is subsequently validated against the quantitative data to assess whether the identified causes are consistent with the observed patterns in the dataset.

5.2.1 Root Cause Analysis

The cause-and-effect diagram constructed in the Measure phase identified causes of low UTL across 6 categories: Machine, Method, Material, Manpower, Measurement and Environment. To structure the analysis, the identified causes were categorised according to Montgomery's (2008) distinction between

chance causes and assignable causes, which determines both the nature of the problem and the type of intervention required to address it.

Several of the identified causes were classified as chance causes, representing sources of variation that are embedded in the current process design. As Montgomery (2008) describes, when only chance causes are present, the process is operating at the maximum level that the current design allows, and reducing variation further would require kaikaku-type interventions. At the coil cutting stage, the combination of old equipment and a planning system that does not automatically group orders with regard to coil width represents the most significant chance causes. The planning system controls production sequence automatically, leading to excessive coil changeovers when the same coil width is removed and refitted multiple times within a single production day. To reduce changeover count, operators manually re-sequence production orders. This is compounded by the limited number of available coils in rotation and the fact that market coil widths do not match the article dimensions precisely, which itself forces wider coils to be used than necessary, which drives trim loss in the punching stage. The nesting logic further contributes as a chance cause, as it is an internally built system rather than a vendor-optimised program, meaning that material layout across the sheet is not systematically optimised, as no dedicated nesting vendor is responsible for its configuration and performance. Together, these causes form an interconnected constraint where MTO-driven order variability, planning system limitations and coil width restrictions compound each other, making UTL particularly difficult to control without fundamental intervention.

A separate set of causes were classified as assignable causes, representing identifiable sources of variation that are traceable to specific events or factors. These can in principle be corrected at the operational level (Montgomery, 2008). The most significant assignable cause identified was the operator-dependent variation at the coil levelling stage. Levelling quality was described as highly person-dependent, with differences between operators estimated to account for up to two meters of additional offcut per coil change. This variation is difficult to standardise through conventional training, as the machine knowledge is built exclusively through on-the-job experience with no external training available. Varying coil quality between suppliers and rust damage from improper internal storage were also identified as assignable causes, as both represent deviations from expected input conditions which cause additional coil offcut generation at the cutting stage (Montgomery, 2008).

The operator-dependent variation identified above was further examined against the quantitative data. To test whether operator skill had a measurable effect on offcut length, operator identity was recorded during the coil offcut measurements. The average offcut length did not differ between the operators by a measurable amount, and their identity was therefore excluded in Figure 4.5. Montgomery (2008) notes that when chance causes dominate process variation, the contribution of assignable causes as operator variation may not be clearly detectable in aggregate data. This is consistent with Andersen and Fagerhaug's (2006) observation that in data-poor environments, qualitative accounts tend to reflect what is operationally visible, rather than what is analytically significant, a pattern directly illustrated here: operators attributed variation to levelling skill precisely because that is the aspect of the process most immediately within their control and perception. This mirrors the broader pattern described by Tversky and Kahneman (1974), where the perceived significance of a cause is shaped by how readily it comes to mind rather than by its actual contribution, and in a production environment where process-level data is absent, what comes most readily to mind is invariably what is most physically proximate and personally controllable. The absence of a measurable operator effect indicates instead that coil condition at the point of loading, a chance cause driven by internal handling and storage practices, is a more influential determinant of offcut length than operator skill. However, given the limited sample of 14 measurements across four production days, this interpretation cannot be conclusively established from

the available data alone, and the relative contribution of operator skill versus coil condition remains an open empirical question.

The measurement category was treated separately from the variation causes. The identified causes in this category, including absence of process-level scrap tracking, uncertain bin weighing, missing historical data and design scrap not visible in existing data, do not drive material loss on their own, but instead represent barriers to visibility and process control. Without process-level measurement, the relative contribution of each cause cannot be determined, improvement efforts cannot be prioritised, and interventions cannot be verified. This corresponds to what Shahbazi et al. (2016) describe as fundamental barriers, where companies are aware of total scrap volumes but lack the process-level data needed to link the losses to specific operations or act on them. Andersen and Fagerhaug (2006) note that where process-level data is absent or fragmented, distinguishing between genuine root cause and first-level symptoms becomes difficult. This study provides empirical evidence of how that risk materialises in practice.

5.2.2 Validation Against Data

Before the collected data can serve as an analytical baseline, the Measure phase requires that the measurement system itself is assessed for reliability (Sörqvist & Höglund, 2017). This section addresses that requirement through four analyses. First, the ERP data and manual bin weighing are cross-validated to assess whether the two sources produce a consistent picture of scrap distribution across stations. Second, the absence of blank removal scrap from the ERP system is examined. Third, coil width mismatch is analysed as a quantifiable source of punch scrap. Fourth, the coil offcut data is assessed to determine whether scrap generated as offcuts follows a pattern consistent with chance causes or assignable causes, including the contribution of operator-dependent variation.

Cross-Validation of Scrap Bin Measurements

Cross-validating the ERP data and manual bin weighing is a necessary step before the collected data can serve as an analytical baseline, as the Measure phase requires the measurement system itself to be assessed for reliability (Sörqvist & Höglund, 2017). The ERP data and manual bin weighing cover different time periods, which limits direct comparison of absolute values. However, comparing their relative distributions remains analytically valid for assessing consistency across sources. The ERP data attributed 74% of logged scrap to the punching machine and 26% to the coil cutting machine. The manual bin weighing distributed scrap across three stations: punching machine at 61%, coil cutting machine at 20%, and blank removal station at 19%. The blank removal station is entirely absent from the ERP data, which is consistent with the observation finding that no active scrap measurement existed at that stage. To enable a comparable distribution, the blank removal station was excluded from the manual weighing figures, and the remaining two stations were renormalised. This yields a punching machine share of 75% and a coil cutting machine share of 25%, closely matching the ERP distribution of 74% and 26%, see Table 4.2. The alignment across two independent measurement methods supports the conclusion that the relative scrap contribution of each station is stable, even across different time periods.

Despite this alignment in relative terms, the manual weighing recorded substantially higher absolute scrap volumes than the ERP data for the same two stations. Basu (2011) notes that automated data sources generally provide higher reliability due to their traceability and lower susceptibility to

measurement error, while manual methods introduce the risk of human-dependent variability. This is directly reflected in the data. During the manual weighing period, weighing routines at the punching station were not consistently followed by the operators, a finding that reinforces the interview theme on unreliable measurement methods identified in section 4.3.1. The team leader in charge of the weighing scrap bins at the punch stage explicitly described the current weighing practice as estimation-based rather than measurement-based, noting that bins were weighed at full capacity once, after which the recorded weight was used as a reference. The reference weight was applied in full when the bin was emptied at full capacity and halved when emptied at approximately half capacity, rather than re-weighing at each emptying. This estimation-based approach illustrates the reproducibility limitation described by Montgomery (2008) as fundamental to reliable measurement systems, where the recorded weight was dependent on the operator's assessment of fill level rather than an independent measurement. This gap is consistent with what Shahbazi et al. (2016) describe as an information barrier, and with Schiffauerova and Thomson's (2006) observation that measurement approaches in manufacturing lack the standardised methodology, producing results sensitive to how and when data was collected. Establishing a reliable absolute baseline at the punching stage would therefore require a redesign of the weighing routine rather than a better execution of the current one.

In practice, 75% of all bin emptyings at the punching station occurred when the bin was at approximately full capacity, meaning that the full-capacity reference weight was applied in most cases. This limits the practical impact of the estimation-based approach on the recorded distribution. However, density variation within the scrap bins remains an uncontrolled source of error regardless of the bin's fill level, as loosely packed scrap and densely packed scrap produce different bin masses even when the bin appears full. Where a formal Gauge R&R study is not feasible due to the absence of a defined instrument or stable measurement conditions, convergence across independent sources becomes the primary available basis for establishing confidence in the combined measurement design (Montgomery, 2008; Yin, 2018). The relative alignment across two independent sources, separated in time, therefore provides sufficient basis for treating the proportional scrap distribution as a reliable finding, even where the absolute volumes remain uncertain. The ERP-derived data should therefore be treated as the more traceable of the two sources, while manual weighing serves as a directional complement rather than a verified baseline (Schiffauerova & Thomson, 2006).

Blank Removal Visibility in ERP Dataset

The blank removal station contributed 19% of total scrap in the manual bin weighing yet has no corresponding entry in the ERP system. As identified in section 4.3.2, Quality Engineer 2 explicitly noted that blank removal scrap *"is not included in the nesting scrap, so you don't see it in the data."* The analysis examines why this absence occurs and what it means for the completeness of the scrap baseline.

The punching machine operates according to a nesting program that classifies all material on a sheet into two categories at the programming stage: finished parts and trim loss. For large geometries, the nesting software deliberately leaves small uncut connections between the inner blank and the surrounding panel to maintain sheet stability during processing. The inner blank is then manually separated at the blank removal station, outside the machine's operational scope. The ERP system records only the dimensions of the finished panel as the output of the punching stage. Because no production transaction is triggered at the point of manual separation, the material discarded at the blank removal station has no corresponding record in the system. This is not a case of miscategorisation but of a systematic gap: the ERP's data model does not include a transaction point corresponding to this physical

event. Frohm et al. (2008) note that where physical tasks are performed manually outside the scope of automated systems, the availability of digital production data is correspondingly low, and the blank removal station exhibits both conditions simultaneously: a manual operation positioned downstream of the machine boundary where ERP logging ends.

Datar and Rajan (2017) distinguish between normal spoilage that is systematically built into the production design and abnormal spoilage arising from specific events, noting that without process-level data these two categories cannot be separated. The design scrap generated at the blank removal station is normal spoilage by nature, since every hole panel produces a knocked-out blank as a direct consequence of the product geometry. Yet because it is unrecorded, it cannot be distinguished from abnormal spoilage or quantified independently without physical measurement, leaving the scrap volume at this stage unquantifiable through the existing system of record. Together, these conditions illustrate a specific mechanism through which significant scrap volumes can remain invisible to improvement efforts: when a manual operation falls outside both the automation boundary and the ERP logging, its material losses are neither logged automatically nor categorised incorrectly, they simply do not exist within the system of record. Establishing visibility at such stages therefore requires active measurement design rather than extraction from existing data sources.

Coil Width Mismatch Analysis

Coil width mismatch was quantified by comparing each article's width against the coil width used during production, expressed as a percentage of the coil width. As shown in Figure 5.2, a significant mismatch exists across all five coil widths; the fact that this persists in coil widths with over 1000 produced articles indicates that suboptimal width selection is not an isolated event but an embedded characteristic of the production system, consistent with Montgomery's (2008) description of chance causes as sources of variation inherent to the process design.

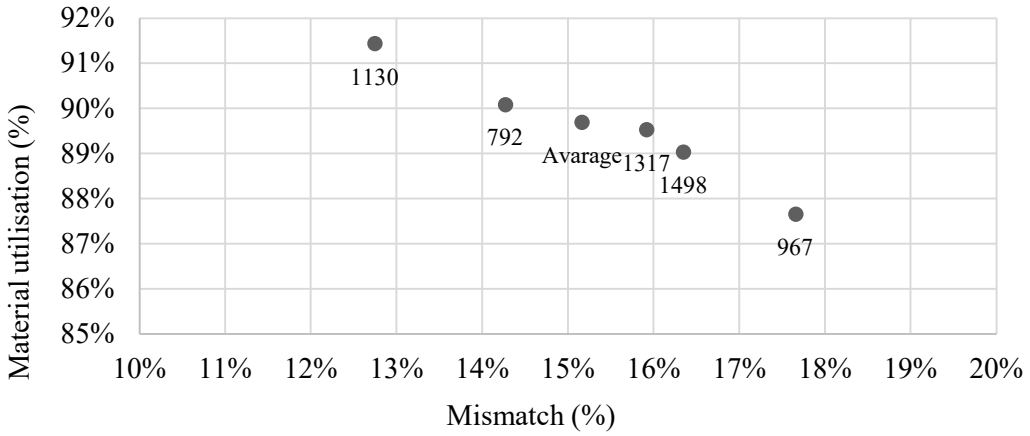


Figure 5.2

Average material utilisation, plotted against mismatch percentage for each coil width

Comment. Coil widths expressed in mm

In the cutting stock typology established by Wäscher et al. (2007), trim loss minimisation presupposes that appropriately sized input material is available. Herrmann and Delalio (2001) confirm this in an NC punch press context specifically, establishing that trim loss is an embedded outcome of the nesting layout

for a given sheet width, meaning that when coil width exceeds article width the material loss is determined before the nesting algorithm even runs. The observed average mismatch of 15%, and the presence of mismatch exceeding 10% in three of the five most frequently produced articles, is a direct empirical expression of this constraint. This is further compounded by the MTO context, as Olhager (2003) and Soman et al. (2004) describe, that article dimensions vary between orders, meaning that the mismatch is not a static issue that can be resolved once, but a continuously shifting constraint that varies with every new order entering the system. The resulting losses correspond to what Datar and Rajan (2017) classify as normal spoilage, which is material losses that persist under efficient operating conditions and can only be reduced through kaikaku-type interventions (Holweg et al., 2018).

Coil Offcut Measurements

Coil offcut measurements presented in section 4.4.2 confirmed that offcut scrap was generated at every change without exception. Juran (1986) describes this type of persistent waste as normal spoilage, which can only be addressed through kaikaku-type interventions rather than corrections (Holweg et al., 2018). Offcut scrap at coil changeover is precisely this type of loss: a consequence of the condition of the first layer of a coil, which must be removed in order for the surface quality to be sufficient for article production regardless of operator or process settings. As Datar and Rajan (2017) describe, losses of this nature persist even under efficient operating conditions and cannot be reduced to zero with the current process design.

The offcut length per coil change is stable across the measurement period, with no systematic trends or shift visible in the data. This pattern is consistent with a process governed by chance causes, where variation is present but not attributable to specific factors (Montgomery, 2008). The practical implication is that improvement efforts should target the frequency of coil changes and the average offcut length per change rather than the elimination of offcut scrap itself. Reducing changeover frequency through improved order sequencing represents a kaizen-type improvement within the existing process structure, while reducing offcut length would require a kaikaku-type intervention targeting the equipment or coil handling design, as explored further in section 5.2 (Holweg et al., 2018).

5.3 RQ3: How can process-level measurement and analysis support the identification and proposal of scrap reduction measures?

RQ3 is addressed through the Improve phase presented in this section. Building on root causes and measurement findings established in section 5.2, four scrap reduction proposals were developed and assessed. Each proposal was evaluated in terms of its targeted scrap mechanism, intervention type, and the degree to which its effect on UTL could be quantified with the available data.

Optimisation of Coil Width Logic

The coil width mismatch analysis presented in section 5.2.2 established that dimensional mismatch between coil width and article width arises systematically from the coil width selection logic, and as seen in Figure 5.2, a clear negative relationship exists between mismatch percentage and UTL. To assess reduction potential, each article was evaluated against all available coil widths to identify whether a narrower option existed that still met the article's minimum dimensional requirement. Articles with a mismatch exceeding 50% were excluded as these represents panels produced multiple per sheet rather than genuine coil-to-article mismatch, excluding 3 161 production records in total.

Of the 25 053 production records included in the analysis, 11 745 were found to have improvement potential under the new selection logic. This corresponds to 2 405 of 5 087 unique article models, indicating that suboptimal coil width selection is not limited to a small subset of articles but distributed across nearly half the range of articles. The current average mismatch percentage across all included records was 12%, which would be reduced to 7% if the optimised selection logic was implemented, which represents a 5-percentage point mismatch reduction. Figure 5.3 illustrates the current and optimised average mismatch per coil width, showing improvement potential across all width categories apart from the 792mm coil, where no narrower alternative exists within the available width range.

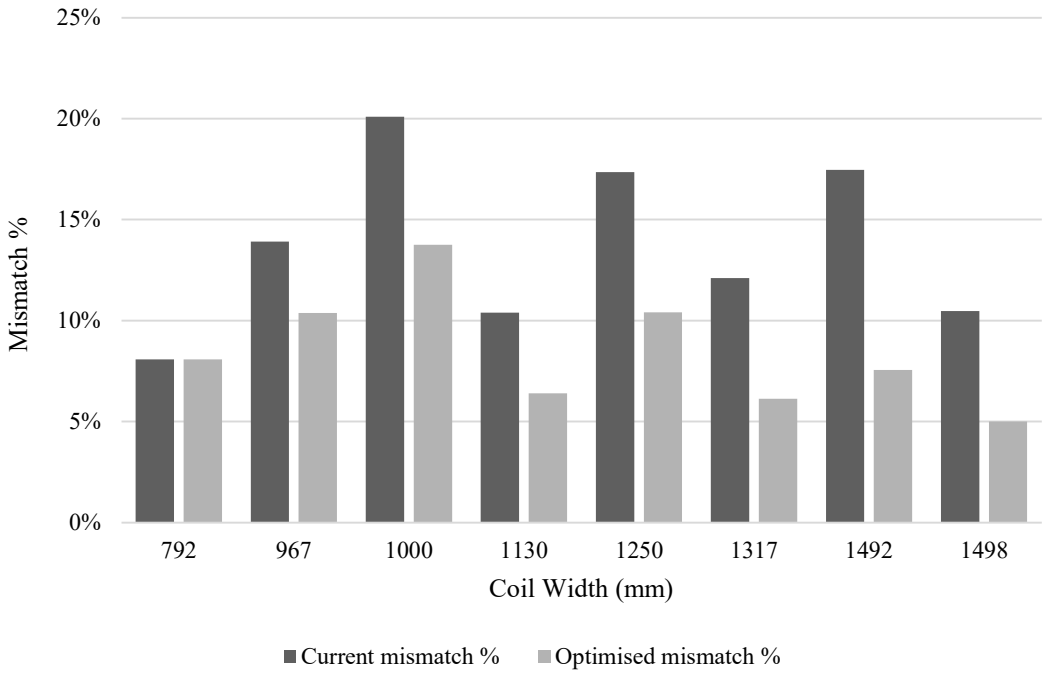


Figure 5.3
Current and optimised coil-width mismatch percentage by coil width (mm)

Among articles produced 50 or more times, the current average mismatch is 7%, reducing to 5% with the implementation of optimised logic. The smaller absolute improvement for high-volume articles reflects that frequently produced articles tend to be better matched to available coil widths or are excluded from the optimisation because they are produced multiple per sheet. Despite this, two of the five most frequently produced articles in the dataset exhibit mismatch levels of 14 and 17% respectively; both would drop below 3% under the optimised coil width logic. Given their high production frequency, even a small mismatch reduction per unit would translate to substantial material savings at scale.

As established in section 5.1.2, the current coil selection is classified as a chance cause, meaning that variation is embedded in the process rather than specific operational events (Montgomery, 2008). The proposed optimisation does not eliminate this cause but reduces the magnitude through better coil width matching within the available range, corresponding to a kaizen-type improvement within the current process constraint (Holweg et al., 2018). The 5-percentage point mismatch reduction represents the upper bound of the expected improvement to total punch UTL, as the actual effect considers the article geometry as well as the mismatch between the article and coil width, which is not captured in the available data. The proposal requires no capital investment and is implementable through a revision of

the coil selection logic in production planning, with the only potential trade-off being a marginal increase in coil changeover frequency if articles are redistributed across coil widths. Analysis of the changeover distribution across coil widths confirmed that this effect remains marginal in practice.

Implementation of 2-Day Batching Logic

The coil offcut analysis established that changeover offcut scrap is generated at every coil change without exception, and that its volume per change is driven by material condition rather than operator skill. The practical consequence is that the total scrap produced in the coil cutting stage over a given period is directly proportional to the number of coil changes performed. Reducing changeover frequency therefore is the most appropriate way to reduce this scrap category within the current process design. As established in section 5.1.2, the planning system generally does not sequence production orders by coil width, meaning that the same width can be removed and refitted multiple times within a single production day. The operators mitigate this through manual re-sequencing; however, this practice is not systematically enforced. Production records were reanalysed using a 2-day batching logic, grouping consecutive production days into pairs and calculating the minimum number of coil changes required within each period. This eliminates redundant width changes that occur only because the same coil width is treated as a separate batch on consecutive days. Implementing 2-day batching reduces total coil offcut scrap by 24% relative to the current baseline. Shingo (1985) argues that when changeover costs cannot be eliminated, the rational operational response is to minimise changeover frequency by grouping orders that share the same material specification, and the 24% reduction estimated represents the gain achievable through a 2-day batching window, though expanding the batching period would theoretically reduce changeover frequency and associated scrap further still, with the practical ceiling determined by the WIP buffer capacity and scheduling flexibility rather than by the batching logic itself. The proposal requires no capital investment and is classifiable as a kaizen-type improvement within the existing process structure, though its feasibility depends on the capacity of the downstream automatic storage system to absorb higher WIP volumes, a dependency that fell outside the scope of this study.

Replacement of Equipment

The empirical findings identified two limitations at the coil cutting stage that machine replacement would address. The first is the rollback mechanism required during coil width changes. As directly observed during the on-site observation sessions, the partially used coil must be manually reversed so that the tensioning wheels, which press down on top of the coil during operation, can be disengaged before the coil is lifted off by a forklift. This reverse motion drags the coil surface against the tensioning wheels under contact pressure, creating surface scratches on the visible side of the remaining material. When that coil is later remounted in a subsequent production run, the scratched part of the coil needs to be cut off and discarded before quality is sufficient for production, adding directly to changeover offcut scrap. Montgomery (2008) classifies this as an assignable cause tied to the specific physical configuration of the current machine, meaning it cannot be resolved through operator training or procedure changes alone, but requires a change to the equipment or process design. The rollback mechanism represents the type of internal setup activity that Shingo (1985) identifies as the highest-priority target for changeover cost reduction, a physical step embedded in every coil width change that generates material loss as a direct and repeatable consequence of the current machine design. That this loss is systematic rather than incidental is supported by the manual offcut measurement data, which confirmed that the offcut length produced by scratches was present at every coil change.

The second limitation is the absence of process data traceability. Although offcut lengths were manually measured during this study, no automated records exist of the underlying process parameters, such as leveller settings, incoming coil condition, or operator actions at each changeover. The consequence is visible in the collected data itself: changeover offcut lengths ranged from approximately 80% to 125% of the average across 14 recorded events, yet the causes of this variation cannot be investigated because the process conditions that produced each measurement were never captured. Frohm et al. (2008) directly link automation level to data availability, noting that manual and semi-automated operations rarely generate the digital records needed for process-level analysis, and the current coil cutting setup is a direct expression of this relationship. Without this contextual data, the variation observed in offcut lengths cannot be attributed to specific causes, meaning the distinction between chance causes and assignable causes that Montgomery (2008) identifies as fundamental to process improvement cannot be made. The result is that the coil cutting stage remains analytically opaque: scrap is generated, measured in aggregate, but its drivers are invisible. A replacement machine with higher automation would close this gap, transforming offcut scrap from an estimated byproduct into a continuously logged and traceable process output, and in doing so create the precondition for applying the full DMAIC analytical cycle to a stage that this study could only partially address.

The intervention carries a higher implementation threshold than the other proposals, requiring capital investment, a production shutdown during installation, and operator retraining. The coil cutting station contributes the smallest share of total system scrap, and this should be considered when prioritising between the proposals. Within the improvement typology described by Holweg et al. (2018), machine replacement represents a kaikaku-type intervention, addressing process-level limitations that the kaizen proposals cannot resolve within the existing process design.

Implementing Measurement and the Reuse of Hole Panel Blank

The blank removal station contributed 19% of total measured scrap by weight yet generated no corresponding record in the ERP system, as established in section 5.1.2. Ingarao (2017) notes that material losses in sheet metal manufacturing arising from part geometry are among the most persistent sources of yield loss precisely because they are embedded in the product design rather than in the process itself, making them invisible to conventional scrap reporting. Establishing visibility of this loss is therefore the analytical prerequisite for any further improvement at this stage. One approach to closing this measurement gap is to supplement the existing ERP data with geometric data drawn directly from the panel drawings. The ERP currently records only the outer dimensions of each panel, but the net finished area, that is the outer panel area minus the internal cutout, can be derived from the technical drawings available for each panel variant. By calculating the difference between gross and net area per article and combining it with material surface density and production volume, an implied blank weight per production run can be established. This is methodologically consistent with the ERP-based derivation already applied at the coil cutting stage in this study, where physical measurements were combined with ERP production counts to estimate scrap weight. The additional effort required lies primarily in extracting and processing the cutout geometry from the drawings for each hole panel variant, which is a one-time calculation task rather than an ongoing measurement burden. Once established, this calculation layer would transform the blank from an untracked byproduct into a documented material output with known geometry, weight and production frequency, providing the empirical foundation that currently does not exist at this stage.

Beyond measurement visibility, the blank also represents a physical recovery opportunity that the current absence of data makes impossible to evaluate systematically. Datar and Rajan (2017) distinguish

between scrap that is built into product design and scrap that arises from process behaviour, noting that the former cannot be reduced through process control alone but requires intervention at the design or planning level. The knocked-out blank falls into this category, as its existence is determined by the AHU cutout geometry rather than by how the machine operates, meaning its status as waste or recoverable material is fundamentally a planning decision. As illustrated in Figure 4.2, the internal cutout occupies a substantial share of the total panel area, making physical reuse a realistic consideration. Without documented blank dimension, surface quality and production frequency across hole panel variants, an in-depth evaluation of physical reuse is not possible with the data available in this study. Of the four proposals, this represents the lowest implementation threshold, requiring a development effort to add article cutout calculations for ERP logging, with no physical changes to the production process. Establishing this visibility is, however, a prerequisite for any further improvements at this station.

5.3.1 Combined Improvement Potential

The four improvement proposals presented in section 5.3 address scrap generation through different mechanisms and at different points in the production system. Table 5.1 summarises the proposals, their targeted scrap mechanism, intervention type and the degree to which their effect on UTL can be quantified with the data available in this study.

Table 5.1
Summary of improvement proposals and their characteristics

Proposal	Station	Mechanism	Type	Quantified
2-day batch	Coil cutter	Reduces changeover frequency	Kaizen	Yes
Coil width optimisation	Punch	Reduces width mismatch	Kaizen	Partial
Machine replacement	Coil cutter	Reduces offcut variation and handling damage	Kaikaku	No
Measurement and reuse of hole panel blank	Blank removal	Establishes measurement and recovery potential	Kaizen	No

Of the four proposals, only 2-day batching can be fully quantified from available data, reducing coil offcut scrap by 24% and increasing total UTL by 1 percentage point. Coil width optimisation is partially quantifiable, but the full effect requires more detailed article geometry data. Across the four proposals, coil width optimisation represents the highest priority for immediate implementation. It targets the punching stage, which contributes the largest share of total system scrap, requires no capital investment, and its effect is partially quantifiable from existing data. The 2-day batching proposal is similarly implementable without capital investment, and its full effect is quantifiable, making it the second logical priority. Together, these two kaizen proposals address the two largest scrap contributors in the system and can be implemented within the existing process design, consistent with Holweg et al.'s (2018)

argument that kaizen interventions should precede kaikaku where incremental improvements remain achievable.

Machine replacement carries a higher implementation threshold and addresses a small contributor in absolute terms. However, its relevance extends beyond scrap production: as established in section 5.2.1, the current machine generates data invisibility at the coil cutting stage that cannot be resolved through process changes alone. This positions machine replacement as a longer-term prerequisite for applying the full DMAIC cycle to that stage rather than an immediate scrap reduction measure. The measurement and reuse proposal at the blank removal station represents the lowest implementation threshold of the four, requiring only a calculation layer added to the ERP system, and is a logical first step since measurement visibility is a precondition for any further improvement at that stage.

6 Discussion

The findings of this study contribute to the theoretical understanding of the DMAIC framework's implementation in manufacturing contexts and the practical challenge of establishing process-level scrap measurement in MTO production. This chapter discusses the implications of the findings of the study on existing theory and practice. It also reflects on the methodological choices and their limitations.

6.1 Discussion of Findings

The findings of this study cannot be understood independently of the process mapping conducted during the Define phase. In production environments where scrap is not yet measured at the process-level, the system boundaries and scrap-generating stages cannot be assumed. The SIPOC diagram and process flow mapping were not preparatory steps carried out before the real work began, but the foundation that made all subsequent phases of DMAIC possible. Without a clear process boundary, the Measure phase would have had no defined scope, and the scrap data collected would have been impossible to interpret. This reflects a broader condition identified in the research gap, where the Define phase must do more work than it typically does in stable, data-rich environments when process-level measurement does not previously exist.

The process mapping revealed that the distribution of scrap-generating stages was not predictable from a general understanding of sheet metal production alone. Stages that are positioned downstream and manually operated are at particular risk of being overlooked, both because they are less visible in automated data systems and because they fall outside the intuitive mental model of where material loss occurs. This finding suggests that in hybrid production systems, process mapping must extend to the full material flow rather than stopping at the boundary of the automated segment, since low-automated downstream stages may carry a disproportionate and previously unmeasured scrap contribution.

The analysis produced three categories of findings. First, it confirmed that the dominant source of scrap in this type of production system is built into the process design, arising from the mismatch between available material dimensions and order-specific requirements rather than from process failures or operator error. Second, it revealed that manual operations at the boundary of the ERP system represent a systematic blind spot in scrap measurement, one that is not captured by automated data collection regardless of how well the upstream logging is designed. Third, it demonstrated that in the absence of process-level data, the intuitive picture held by production personnel reflects physical proximity to scrap rather than actual scrap volumes, making it an unreliable foundation for improvement prioritisation.

The confirmation of coil width mismatch as the primary root cause of low UTL was consistent with what the theoretical framework had predicted. In an MTO environment where article dimensions vary rapidly between orders and the available coil width selection cannot perfectly match every article, some degree of mismatch is expected as normal spoilage inherent to the production system, rather than a sign of process failure (Datar & Rajan, 2017; Olhager, 2003). What was not anticipated was that systematic mismatch persisted even among the highest-frequency articles, where the production volume is sufficient to justify targeted coil width procurement. This suggests that in MTO environments without active UTL monitoring at the article level, even correctable sources of normal spoilage can remain invisible and therefore unaddressed indefinitely, regardless of production volume. Had process-level utilisation data been routinely monitored, systematic mismatch of this amount across high-volume articles would have surfaced as an apparent pattern, prompting a corrective action. The finding therefore reinforces the central argument of this study: that the absence of process-level measurement does not

merely make improvement efforts harder to execute; it prevents the identification of improvement opportunities that would otherwise be invisible.

The blank removal station finding illustrates a broader systematic risk in hybrid production systems. When the boundary of the automated data collection system does not extend to the full length of the production flow, the stages that fall outside it become invisible contributors to total scrap volume. This is not a measurement error in the conventional sense but a consequence of how ERP systems are typically implemented around automated operations, leaving manual downstream steps unrecorded by design. This kind of invisible scrap is precisely what Shahbazi et al. (2016) describe as an information barrier, and the finding confirms that it can represent a material share of total scrap production even when upstream measurement is functioning well. Addressing it requires deliberate expansion of the measurement scope rather than refinement of the existing data collection approach.

The discrepancy between production personnel intuition and the measured scrap distribution was the finding that most directly invalidated a pre-study assumption. It had been expected that interviews and observations would provide a reliable and in-depth qualitative picture that would develop into a holistic understanding of the scrap production within the production process. Instead, the qualitative picture turned out to be systematically biased in a direction that could be explained by physical proximity to the scrap rather than by the actual scrap-producing actions of the process. This is consistent with the availability heuristic described by Tversky and Kahneman (1974), where the perceived frequency of events is shaped by how easily instances come to mind. In contexts where certain stages require operators to physically handle and separate scrap, those stages become cognitively prominent regardless of their actual contribution to total scrap volume. The practical consequence is that in environments lacking process-level data, the subjective picture held by personnel closest to the process may be unreliable as a basis for improvement prioritisation. Efforts directed by this kind of intuition rather than data tend to target what is visible rather than what is significant, and without an implemented measurement system to provide feedback data, there is no mechanism for monitoring and correcting that bias over time.

A broader pattern emerges when considering the measurement challenges encountered in this study. The most significant measurement uncertainties, namely the absence of automated logging at the coil cutting stage, and the inability to fully verify the weighing routine are not incidental methodological shortcomings, but direct consequences of the measurement barriers the study set out to investigate. The coil cutting stage lacked automated data collection because it operated below the automation threshold at which digital records are produced, consistent with what Frohm et al. (2008) describe as the relationship between automation level and data availability. Together, these constraints confirm that measurement limitations in low-automation MTO environments are not incidental shortcomings but inherent features of the production context, and that resolving them requires changes to how measurement is designed rather than how it is executed. The conditions that produced this blind spot, namely ERP systems implemented around automated stages with manual downstream steps excluded by design, are common in hybrid MTO sheet metal production. This supports the analytical generalisability of this finding beyond the single case.

Taken together, the findings can be summarised around three analytical threads. The three scrap flows, coil changeover offcut, trim loss and blank removal scrap, were all classified as normal spoilage, the four interview themes spanning measurement challenges, scrap sources, variation causes and improvement opportunities confirmed the qualitative picture, run charts and histograms established that UTL variation is stable and governed by chance causes. Together, these findings provided the empirical

foundation for four targeted improvement proposals: coil width optimisation, 2-day batching, equipment replacement and blank removal measurement.

6.2 Theoretical Implications

A recurring assumption in the DMAIC literature is that the Measure phase serves to quantify a problem whose existence and location is already known. Existing case studies demonstrating scrap reduction through DMAIC have largely been conducted in environments where process-level measurement data was available prior to the improvement effort (Enache et al., 2023; Muhammad Hamad et al., 2021). This study tests that assumption in a context where it does not hold, and the findings both confirm and extend what the theoretical framework predicted.

The relationship between automation level and data availability, as described by Frohm et al. (2008), was directly observable in this production system. Stages operating below the threshold of information automation generated no digital records, meaning that the data required for process-level analysis had to be constructed rather than extracted. This is consistent with what Shahbazi et al. (2016) describe as an information barrier to material efficiency: the absence of structured data collection at manual operations is not an oversight but an inherent feature of how ERP systems are typically implemented around automated processes. The implication for DMAIC theory is that in hybrid MTO production systems, where automation levels vary across the production flow (a condition Holweg et al. (2018) describe as characteristic of batch-and-line configurations), the Measure phase cannot assume that a measurement system exists. It must treat measurement design as an active output rather than a precondition (Sörqvist & Höglund, 2017). This is not a claim about DMAIC in general but specifically about its application in contexts where automation levels vary across the production flow and measurement infrastructure cannot be assumed to exist at the outset.

The findings also extend the theoretical understanding of normal spoilage in MTO production. While Datar and Rajan (2017) define normal spoilage as waste expected under efficient operating conditions, the coil width mismatch finding suggests that in practice this boundary is not always clearly drawn. Spoilage that could be reduced through targeted procurement decisions had been implicitly treated as unavoidable, because the absence of process-level monitoring meant the pattern was never visible enough to prompt a response. This points to a gap in how normal spoilage is operationalised in MTO environments: without active UTL monitoring, the distinction between genuinely unavoidable spoilage and correctable spoilage that has not yet been acted upon cannot be made.

6.3 Practical Implications

The central practical implication of this study is that process-level scrap measurement is achievable under conditions of limited automation and high variability in product mix, but achieving it requires a measurement approach designed around the specific structure and data availability of the production system. The combination of ERP extraction and targeted manual weighing demonstrated in this study offers a transferable starting point for establishing a measurement baseline where none previously existed. However, the value of this combination depends critically on cross-validation between the two sources, as neither is sufficient in isolation and the limitations of each are not visible from within the source itself.

A second practical implication concerns the data extraction process specifically. Establishing the ERP-based component of the measurement baseline was not a straightforward technical step but an iterative process that required close collaboration with an internal contact who combined direct ERP system access with sufficient production knowledge to interpret what the data represented physically. This collaboration was integral to identifying which fields were relevant, understanding what the logged data actually captured in terms of physical events, and recognising where the data did not correspond to what happened on the production floor. For similar companies considering a process-level measurement initiative, this represents a real practical threshold: accessing and interpreting ERP data requires someone who can bridge the gap between system knowledge and production understanding. This is not necessarily a barrier, but it is a dependency that needs to be planned into the project, and it suggests that such initiatives are best pursued as a collaboration between researchers or analysts and internal contacts with operational knowledge.

A third practical implication follows from the finding that aggregate plant-level scrap reporting actively conceals the distribution of losses across process stages. For MTO sheet metal manufacturers more broadly, this suggests that UTL at the process stage should be an operational Key Performance Indicator (KPI) rather than a research output. The punching stage in this study already generates nesting UTL data as a by-product of the batching process, meaning that the metric is latent in the existing infrastructure and requires only consistent extraction and monitoring to become useful. The barrier is not technical but organisational: without a deliberate decision to track and act on process-level data, the information remains unused and improvement efforts are misguided by aggregate figures.

A fourth implication concerns the sustainability dimension of process-level scrap measurement. Material scrap in sheet metal manufacturing represents not only a financial loss but also a consumption of raw material that has already generated emissions through extraction, processing, and transport. As established in the introduction, reducing material losses in the metal industry is directly linked to reductions in CO_2 emissions and resource extraction (Milford et al., 2011; Shahbazi et al., 2016). The improvement proposals developed in this study, particularly coil width optimisation and 2-day batching reduce the volume of material scrapped per unit of output without requiring additional energy input or capital investment. In this sense, the measurement approach developed here serves as an operational improvement and a prerequisite for making the environmental cost of material waste visible at the process-level. Without process-level measurement, the sustainability implications of scrap generation remain as invisible as the scrap itself.

6.4 Contributions

The study makes two main contributions. Theoretically, the study contributes to the applied DMAIC literature by providing an empirical account of implementing MSA in a hybrid MTO sheet metal environment where no process-level scrap baseline previously existed. Where existing DMAIC research largely applies the framework to processes with established digital measurement infrastructure, this study demonstrates how a valid measurement system can be constructed when information automation is absent at key production stages, by combining ERP extraction at automated stages with targeted manual weighing at manual ones (Frohm et al., 2008). The central empirical finding supporting this contribution is the identification of 19% of total weighed scrap at the blank removal station, a volume that was entirely absent from the enterprise system and would have remained undetected without the manual weighing component of the measurement design. This finding also provides concrete evidence of how the availability heuristic (Tversky & Kahneman, 1974) can misguide improvement prioritisation

in the absence of process-level data. Production personnel consistently attributed the majority of scrap generation to the coil cutting stage, which was the stage where scrap was most physically visible, rather than to the actual distribution across the process.

The second contribution is practical. The mixed-method measurement approach demonstrated in this study offers a transferable model for MTO sheet metal manufacturers operating with hybrid process structures and limited automation at downstream stages. For such manufacturers, the principal barrier to establishing process-level UTL as an operational KPI is not technical but organisational: the data required is often latent in existing infrastructure and requires only deliberate extraction and monitoring to become actionable. The study demonstrates that this is achievable under conditions of limited automation and high product variability, provided the measurement approach is explicitly designed around the actual data availability of the production system rather than assumed to be in place. The transferability claim rests on analytical rather than statistical grounds. The conditions of MTO production, hybrid process structure and limited automation documented in this case are not unique to this facility and have been identified as a broad feature of this manufacturing category by Frohm et al. (2008) and Shahbazi et al. (2016).

6.5 Methods Limitations

The study's findings are bounded by limitations related to both data quality and study design. The qualitative data, comprising three observation sessions and four interviews through purposive sampling, provided sufficient overview of the production flow but represented process snapshots rather than a comprehensive picture of production behaviour. As discussed in section 6.1, interview-based accounts of scrap distribution were systematically biased toward visibly handled scrap rather than its actual distribution, a pattern that reinforces the study's central argument about unreliability of intuition-based scrap assessment in the absence of process-level data. Regarding the quantitative data, the coil cutting scrap figure relied on two assumptions that were reasonable but not independently verifiable: that the measured offcut lengths are representative for the full measurement period, and that the calculated coil change count accurately reflected the actual changeovers performed. Manual weighing similarly carried human-dependent variability, though its proportional alignment with the ERP-derived distribution reduces the likelihood that the relative findings are affected.

These data limitations are compounded by the constraints of the study design itself. As a single case study conducted over one academic semester, the depth and duration of the data collection was time constrained, and the findings are analytically rather than statistically generalisable. The specific scrap and process parameters are context-dependent, but the measurement approach and the identified relationship between automation level and measurement system design are transferable to manufacturers operating in MTO sheet metal production with limited data availability, as discussed further in section 6.3.

7 Conclusions

This study applied DMAIC as a structured approach for identifying and analysing process-level scrap generation in a hybrid MTO sheet metal production environment where no measurement baseline previously existed. DMAIC proved well suited to this context, but its application revealed a challenge specific to low-automation MTO environments. The Measure phase cannot proceed as it typically does in the existing literature, because the measurement capability that the framework presumes was not in place. Establishing that capability was therefore considered an output of the Measure phase rather than a precondition for it. The three conclusions below follow directly from that finding.

7.1 RQ1: How can material scrap and UTL be defined and measured at process-level within an MTO sheet metal production system?

Process-level scrap measurement is achievable in hybrid MTO sheet metal production under conditions of limited automation, but only when the measurement approach is explicitly designed around actual data availability rather than assumed to be in place. In this case, a combined approach using ERP extraction at the punching stage and manual scrap bin weighing at the coil cutting and blank removal stages was required, as neither source was sufficient in isolation. The two sources produced consistent relative scrap distributions across the stages they both covered, confirming that a reliable process-level baseline can be constructed through cross-validation of independent sources. The ability to cross-validate the two sources depended on their overlap at the punching stage, which suggests that a shared measurement point is a useful design condition when combining manual and automated data collection.

7.2 RQ2: What challenges and sources of variation affect the measurement and control of material scrap in the scrap generating stages of an MTO sheet metal production system?

The dominant measurement challenge was the absence of scrap data at stages operating below the automation threshold at which digital records are produced, most notably the blank removal station, which generated 19% of the weighed scrap but had no ERP record. In addition, subjective accounts from production personnel were systematically biased towards visibly handled scrap rather than its actual distribution, limiting their usefulness for determining scrap distribution data. The dominant sources of variation were chance causes embedded in the process design: coil width restrictions, planning system limitations and the age of the coil cutting equipment. Among the assignable causes identified, coil condition at the point of loading, driven by variation in supplier quality and damage from internal storage, appeared to be the primary driver of variation in scrap generation at the coil cutting stage.

7.3 RQ3: How can process-level measurement and analysis support the identification and proposal of scrap reduction measures?

Process-level measurement enabled the identification of four targeted proposals tied to verified root causes rather than general improvement intuition: 2-day batching, coil width optimisation and blank removal measurement as kaizen-type interventions, and equipment replacement as kaikaku-type interventions. More fundamentally, this study demonstrates that in production contexts where process-

level data did not previously exist, the Measure phase of DMAIC should be treated as a phase whose primary output is the measurement system itself. Without that capability, the specific improvement opportunities identified in this study would have remained invisible.

7.4 Future Research

This study completed the Define, Measure, and Analyse phases of the DMAIC framework and proposed improvement measures in the Improve phase. The Control phase was excluded due to the time constraints of the thesis project. Future research should therefore extend this work by implementing the proposed improvement measures and establishing a formal control system to sustain them. This includes defining control charts for the identified key metrics at both the coil cutting and punching stages, assigning clear process ownership, and setting thresholds that trigger corrective actions when UTL deviates beyond expected limits. Evaluating the actual outcomes of the improvements against the estimated potentials quantified in this study would validate whether process-level measurement can serve as a reliable foundation for structured scrap reduction in MTO sheet metal production.

A second direction concerns the transferability of the measurement approach developed in this study. The process-level measurement model was designed for an MTO sheet metal environment where scrap data was either absent or only available in aggregate form. Future research could apply this approach in other MTO production contexts with similar characteristics to assess whether the measurement design holds across different industrial settings and contributes to the development of a more generalised framework for process-level scrap measurement in variable production environments.

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Appendices

Appendix A: Observation protocol

Observation place:
Time & date:
Process name:
Observers:

Category	*Observation Point*	*Answer*	*Elaboration*
Process properties	What process steps can be observed?		
	Which process steps can be seen producing scrap?		
	What is the automation level of the process step?		
Flow properties	Is there idle time between steps?		
Scrap Generation	What type of scrap is being produced?		
	Is the scrap being produced continuously or intermittently?		
	Is the scrap being produced in the starting phase or the changeover phase?		
Variation	Are machine settings static?		
	Are machine settings being altered mid run?		
	Can defects be easily identified in the process?		
Measurement	How is the scrap being measured?		
	If not, how can the scrap be measured?		
	Can the scrap production be traced?		

Appendix B: Interview Guide 1

Intervju guide 1: Teamleader		
Q#	**Fråga**	**Svar**
1	Hur länge har du arbetat här på företaget?	
2	Vad har du för arbetsuppgifter som Teamledare?	
3	Vet du om ni aktivt följer eran materialutnyttjande?	
4	I vilket processteg tror du mest skrot produceras?	
5	Har du nån uppskattning över hur många Kg skrot det är som produceras där?	
6	Är det nåt speciellt I det steget som är orsaken till att skrot produceras? Isåfall vad är det?	
7	Kan du ge ett förslag på några åtgärder som hade minskat skrotproduktionen i det processteget?	

Appendix C: Interview Guide 2

Intervju guide 2: Operatör		
Q#	**Fråga**	**Svar**
1	Hur länge har du arbetat som operatör här på företaget?	
2	Vad har du för arbetsuppgifter som operatör?	
3	Vilka kunskaper behöver man för att vara operatör?	
4	Har du nån form av utbildning inom området LEAN?	
	Tema: Skrot	
5	Vilket processteg producerar mest skrot I linen?	
6	Vad händer med produkten I detta steg?	
7	Hur producerar detta processteg skrot?	
8	Kan du påverka mängden skrot som produceras I detta steg?	
9	Isåfall, hur kan ni påverka?	
	Tema: Defekter	
10	Har ni en rutin som definierar vad som räknas som defekt?	
11	Följs denna rutin?	
12	I vilka processteg kan ni identifiera defekter?	

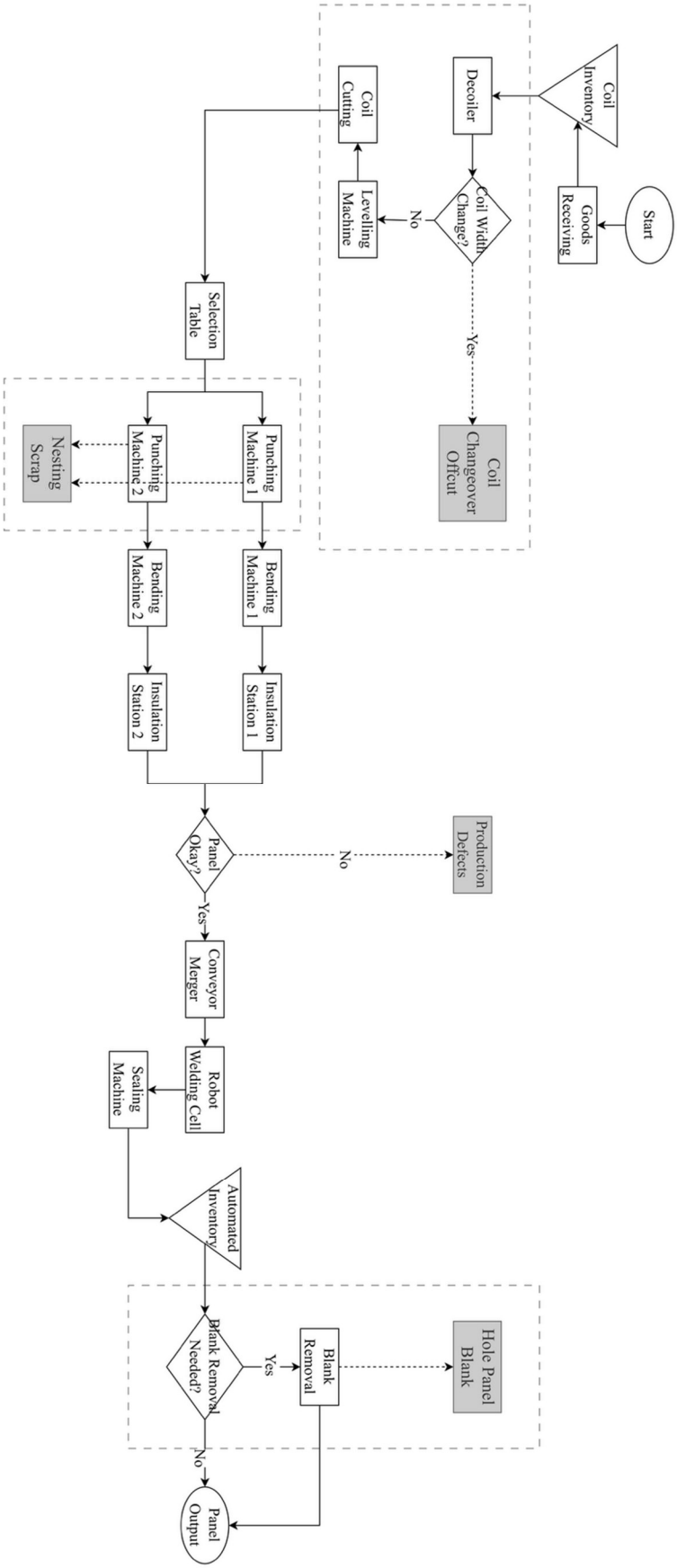
Appendix D: Interview Guide 3

Intervju guide 3: Kvalitet - Processtekniskt		
Q#	**Fråga**	**Svar**
1	Hur ser din karriär ut, från början tills nu?	
2	Vad är din titel och vad har du för arbetsuppgifter?	
3	Vet du om ni aktivt följer eran materialutnyttjande?	
4	I vilket processteg tror du mest skrot produceras?	
5	Är det nåt speciellt I det steget som är orsaken till att skrot produceras? Isåfall vad är det?	
6	Hur kan du påverka direkt materialutnyttjandet?	
	Tema: Processen	
7	Hur produceras skrot i följande processsteg:	
8	Coil monteringen	
9	Coil utrullningen	
10	Riktningsverket	
11	Klippningen	
12	Selection tablen	
13	Stansen	
14	Bockningen	
15	Isoleringen	
16	Svetsningen	
17	Tätningen	
18	BINAR	
19	Slutmonteringen (Panelutplock)	
	Tema: Plåtklippning	
20	Kan du ge ett förslag på några åtgärder som hade minskat skrotproduktionen i STAM:men?	
21	Hade en ökad automationsnivå minskat skrotproduktionen i STAM:men?	
22	Hur är coilbreddarna tillgängliga anpassade till artiklarna?	
23	Finns det anledningar till att de tillgängliga coilbreddarna skulle ses över?	
	Tema: Stans	
24	Kan du ge ett förslag på några åtgärder som hade minskat skrotproduktionen i stansen?	
25	Kan stora hålbilder återanvändas för produktion?	

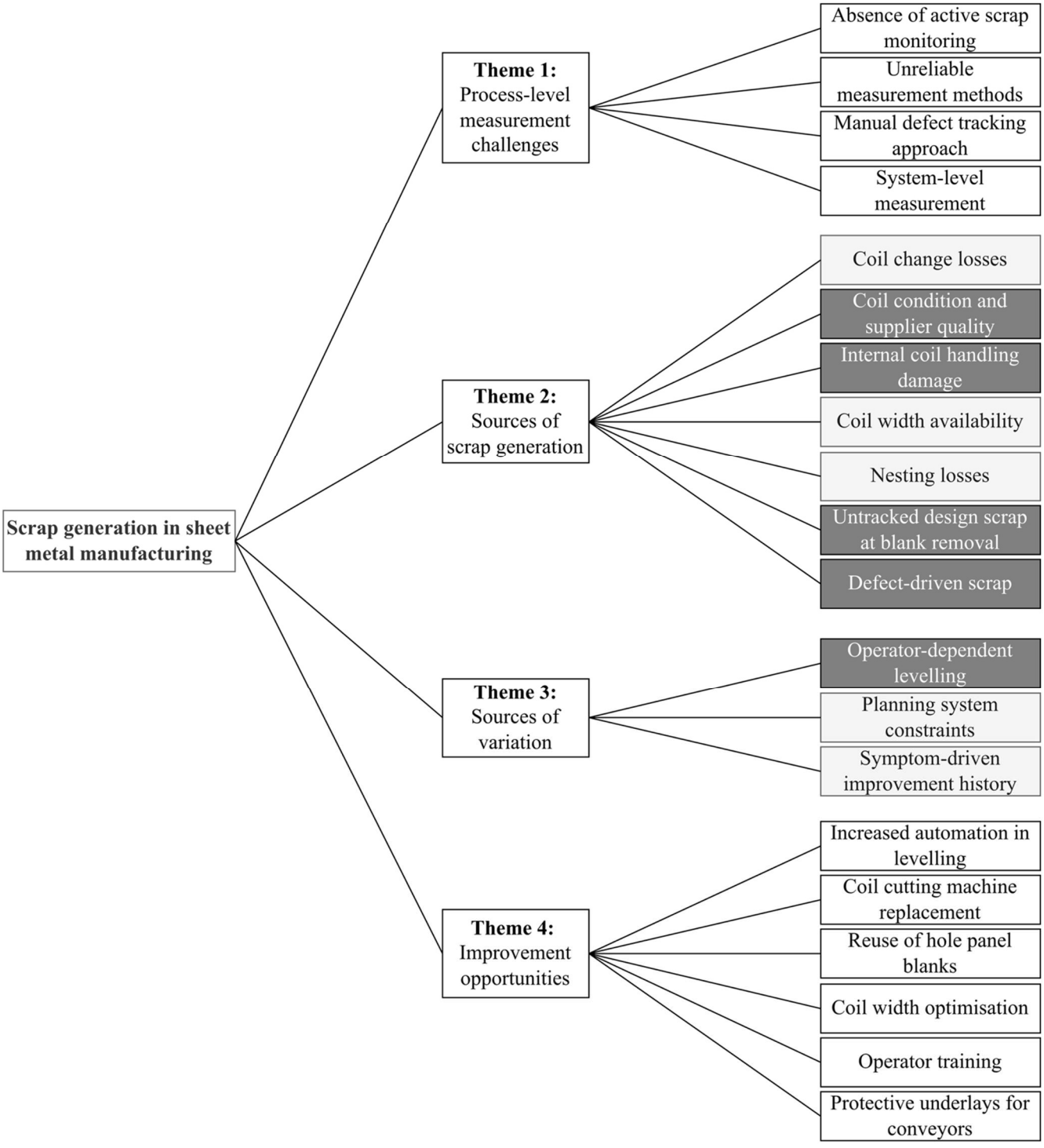
Appendix E: Interview Guide 4

Intervju guide 4: Kvalitet - Defekter		
Q#	**Fråga**	**Svar**
1	Hur ser din karriär ut, från början tills nu?	
2	Vad är din titel och vad har du för arbetsuppgiftter?	
	Tema: Defekter och kassation	
3	Hur definierar ni vad som räknas som defet kontra acceptabelt?	
4	Vad är de vanligaste defekttyperna ni ser från P20?	
5	I vilket processteg fångas flest defekter upp? Och I vilket steg borde de fångas upp?	
6	Hur kan du påverka direkt materialutnyttjandet?	
7	Hur stor del av det kasserade materialet beror på kvalitetsbrister vs. processdesign?	
	Tema: Mätning och uppföljning	
8	Har ni nåt system för att spåra kassationsorsaker?	
9	Hur rapporeras kvalitetsavvkelser idag?	
10	Följer ni upp hur mycket skrot som produceras pga kassering?	
	Tema: Rotorsaksanalys och förbättring	
11	När en defekt upptäcks, finns det en standardprocess för rotorsaksanalys?	
12	Har ni gjort nån rotorsaksanalys kopplad till skrot eller kassation de senaste åren? Vad kom ni fram till?	
13	Finns det några defekter som ni vet om men "accepterar" för att åtgärden anses för dyr eller komplex?	
14	Hur kommuniceras kvalitetsproblem tillbaka till operatörerna?	
	Tema: Förbättringsförslag	
15	Vad är en åtgärd som du tror hade lett till en stor minskning av kassation/defekter om du fick välja fritt?	
16	Finns det nåt produktionssteg där kvalité tycker att skrotnivån är oacceptabel men ingen åtgärd har utförts?	

Appendix F: Process flow diagram



Appendix G: Thematic Analysis tree



Appendix H: Cause-and-effect diagram

